Jump-Off Analysis

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Executive Summary

- PPNR markets models generate scenario-based forecasts of the change in trading desk revenues. To obtain quarterly revenue forecasts in levels for CCAR and QMMF processes, developers use a derived 'jump-off' value instead of actual Q_0 revenue.

- The jump-off value is critical. For first-difference models, adjusting the jump off by $100mm impacts the forecast by $400mm, $900mm, and $1.5bn at 4Q, 9Q, and 15Q, respectively.

- Currently, developers use a decision tree framework that prescribes a 4Q or longer average of trailing revenue. E.g. Q1 2025 forecasts use the average of 2024 revenues. The decision tree and 4Q methodology have critical flaws that materially impact forecasts.

- Objective: redevelop the jump-off framework and adopt a quantitative methodology that prescribes jump-off values per desk. There are three guiding principles:

    - Minimize modeler and business intervention and ad-hoc decision making
    - Preserve economic intuition and time series properties of the data
    - Favor interpretability and usability so the framework can be validated and implemented

- Separate historical revenue into an economically meaningful signal and noise. Noise can encompass idiosyncratic market shocks or business conditions and measurement error, for example. The optimal jump-off is the signal value at Q_0. Smoothing or denoising actual provides an estimate of the signal and permits inferences about the quality of method used.

$$\text{Revenue} = \text{Signal} + \text{Noise}$$

- Using 2010 Q1 – 2024 Q4 revenue actuals for markets desks, this paper evaluates conventional time series models and filters, regressions, and signal processing methods, for the purpose of smoothing or denoising revenue actuals to estimate a jump-off value per desk.

- Assessing signal quality requires an evaluation mechanism. The paper outlines statistical criteria motivated by the signal plus noise framing and desirable properties of a smoothed series.

- Based upon the guiding principles, method scores and additional testing results, I recommend:

    - Complement the 4Q average and decision tree framework with Auto SARIMAX (Seasonal Autoregressive Integrated Moving Average with eXogeneous regressors) and trend/seasonality regression (a regression of revenue, in levels, on quarterly indicators and time trends) models.

    - For each desk, use historical revenues to fit the models, and take the in sample fitted value at Q_0 as the jump-off value. The models initially will serve as business tools. Modelers can update and distribute jump-off values on a quarterly basis.

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Background

PPNR markets models produce scenario-based forecasts for revenue by taking the forecasts from change regressions and mapping them to levels using a point estimate of the jump-off, or Q_0, revenue. For regressions with first-differenced or ratio dependent variables, post-regression derived output is highly sensitivity to the choice of jump-off and the effect can compound over longer horizons[1]:

Jump Off Analytical Sensitivities

Revenue Transformation	Jump-Off Adjustment	Impact on Forecast		
		4Q	9Q	15Q
NoD1 Rev.(t) – Rev.(t-1)	± $100 MM	± $400 MM	± $900 MM	± $1,500 MM
PCT1 Rev.(t)/Rev.(t-1)-1	± X%	± X%		
DP1 Rev.(t)/Rev.(t-1)				
LND1 Ln(Rev.(t)/Rev.(t-1))				

Sensitivity of revenue forecasts, in levels, at 4Q, 9Q, and 15Q, to perturbations to jump off based on revenue model transformation.

A $100mm change to the jump-off value alters revenue forecasts by $1.5bn at a 15-quarter horizon. For desks with modest revenues, such as the Finance Desk, the jump-off is as important as the revenue forecasting model. Ratio transformation revenue models show a static effect. Increasing the jump off by 10%, for example, increases forecasted revenue at all horizons by 10%. Using Q_0 revenue without modification thus exposes forecasts to an aberrational starting point, possibly on account of idiosyncratic market conditions, desk positioning, or noise/estimation error in the data.

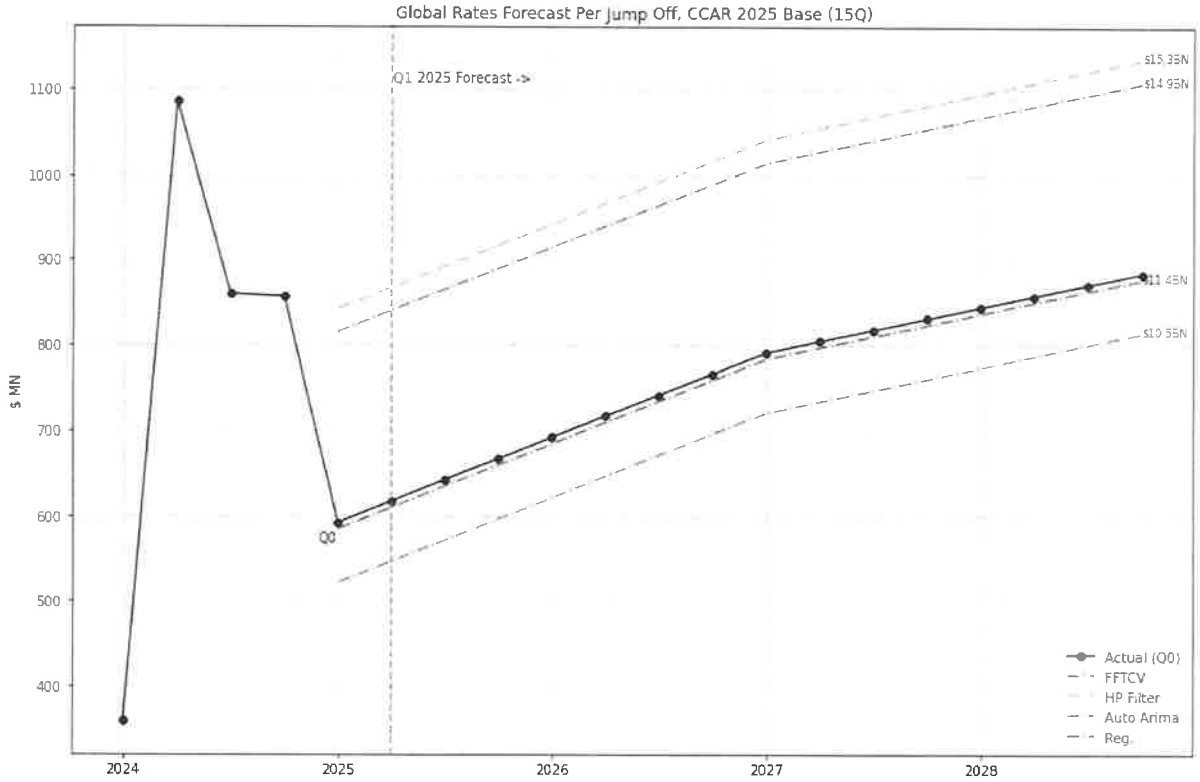
Currently, PPNR modelers use a 4-quarter average of actuals over periods $Q_0\, Q_{-1}\, Q_{-2}\, Q_{-3}$, where forecasts begin at index 1, as part of a broader decision tree approach. Because of deficiencies in this approach, I set out to evaluate quantitative denoising and smoothing methods for Q_0. Each method takes the complete history of observed revenue actuals and returns a fitted Q_0 for the market desks in scope, which include Commodities, DM Credit Trading, DM Structured Products Trading, EMCT, Equity Admin, Equity Cash, Equity Derivatives, F&S, FDC Clearing, FX, Finance Desk, G10 and Local Market Treasuries, GSP, Global Rates, and Prime Finance. I outline guiding principles for method evaluation and selection:

1. Prefer less intervention and ad-hoc decision making by modelers and the business. Calibrate model hyperparameters using cross-validation, or preferably use standard hyperparameters.
2. Preserve economic and statistical intuition. Revenue actuals is a time series of an economic quantity and should be treated accordingly.
3. The chosen method(s) must be practical and robust enough to pass model validation and be used by FP&A, LFOs, and the business. Favor simplicity, interpretability, and usability.

There are countless filtering, smoothing, and denoising methods. It is beyond the scope of this exercise to consider all. I focus on conventional time series filters, regression-based methods, and signal

[1] See the Appendix for the derivation of analytical sensitivity measures. These measures are useful for business planning purposes.

processing techniques, specifically the Hodrick-Prescott and Hamilton Filters, SARIMAX, trend-seasonality regression, Lowess, Wavelet, and Fast Fourier Transform. As baselines, I use moving average and cubic spline smoothing. Historical revenue actuals for each desk are quarterly, spanning January 2010 – December 2024, and non-seasonally adjusted. The figure below demonstrates the materiality of jump-off choice using Global Rates' CCAR Q1 2025 base forecasts using values from selected methods. The jump-off translates the levels forecast up or down, resulting in cumulative revenue $BN apart.



15Q Revenue forecasts in levels ($MN) for Global Rates using CCAR 2025 IBL 1Q scenario. Q0 determined using FFT with cross-validation, HP Filter, Auto ARIMA, and trend/seasonality regression. Cumulative 15Q forecasts annotated at end of plot. Dashed blue line indicates Q1 2025.

The paper proceeds as follows. First, I consider the time series properties of the data used. Second, I evaluate the decision tree currently used to determine Q_0 and highlight deficiencies. Third, I provide a broad signal plus noise framework and for each smoothing or denoising method, background, brief mathematical exposition, and discussion of hyperparameter tuning. Fourth, I develop a novel scoring framework to rank the methods performance on historical data. Finally, I recommend a revised jump-off framework for PPNR markets usage that identifies the best smoothing/denoising methods considered.

Data

Revenue forecasts for the CCAR 2025 cycle begin in Q1 2025. I use a panel of historical total revenue actuals for 17 in-scope PPNR markets desks to calibrate the various smoothing methods and obtain a fitted Q_0 value for Q4 2024. Raw data is monthly and non-seasonally adjusted. I aggregate to quarterly by summing the three monthly observations within a quarter. Data spans Q1 2010 – Q4 2024, inclusive (T=60), covering a long history for each desk that includes a complete interest rate cycle, the Covid shock, and the recent high inflation period. All actuals are reported in $MN. I check for missing observations (there are none) but otherwise do not perform additional data quality checks.

A summary of the time series properties for each desk's historical revenues is below. I use standard tests to assess the presence of seasonality, serial correlation, stationarity, and structural breaks, and establish a point of comparison for the smoothed series. It would be undesirable for a time series of actuals that contains no serial correlation, for example, to display serial correlation when smoothed. Desk revenues in levels uniformly exhibit serial correlation and several are I(1): DMCT, Equity Admin, Equity Cash, Equity Derivatives, Finance Desk, and G10 MT. Auto SARIMA fits for these desks should include a first-difference, and the presence of time-varying moments favors smoothing methods that use local data.

Desk Revenue Time Series Properties

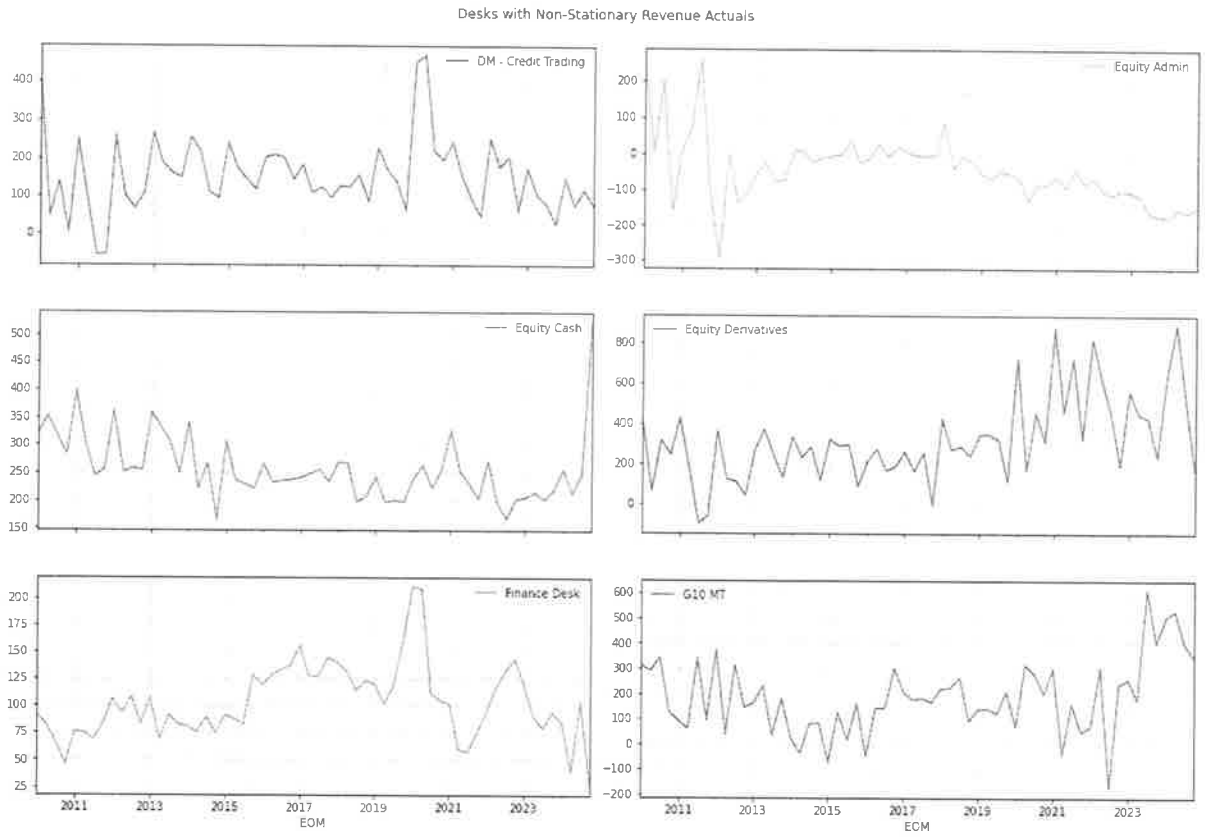
Desk	Seasonality P-Value	Seasonality Present	LBP P-Value	Serial Correlation	ADF P-Value	Stationary	CUSUM P-Value	Structural Break
Commodities	0.640	False	0.000	True	0.000	True	0.929	False
DM - Credit Trading	0.000	True	0.011	True	0.150	False	0.726	False
DM – Struct. Prod.	0.001	True	0.002	True	0.025	True	0.970	False
EMCT	0.000	True	0.000	True	0.036	True	0.958	False
Equity Admin	0.397	False	0.000	True	0.564	False	0.553	False
Equity Cash	0.011	True	0.003	True	0.980	False	0.505	False
Equity Derivatives	0.000	True	0.000	True	0.618	False	0.560	False
F&S	0.551	False	0.000	True	0.001	True	0.281	False
FDC - Clearing	0.924	False	0.000	True	0.000	True	0.560	False
FX	0.008	True	0.004	True	0.000	True	0.199	False
Finance Desk	0.563	False	0.000	True	0.229	False	0.049	True
G10 MT	0.912	False	0.000	True	0.126	False	0.130	False
GSP Other	0.881	False	0.000	True	0.000	True	0.111	False
Global Rates	0.021	True	0.020	True	0.000	True	0.629	False
LMT	0.721	False	0.000	True	0.002	True	0.030	True
Other	0.025	True	0.053	False	0.000	True	0.611	False
Prime Finance	0.630	False	0.000	True	0.000	True	0.121	False

Time series properties of quarterly desk revenue actuals. Actuals span Q1 2010 – Q4 2024, inclusive (T=60), are non-seasonally adjusted, and reported in $MN. Test for seasonality using a Kruskal-Wallis non-parametric test, Stationarity with an ADF test (4 maximum lags, select AIC-minimizing lag), serial correlation with a Ljung-Box test (4 lags), and a CUSUM test for a structural break, using a 5% level for all tests.

To test for a structural break, I use a CUSUM test on the residuals of a time trend regression[2]. Seasonality is present for a majority of desks; hence I expect that for DMCT, EMCT, Equity Derivatives,

[2] A CUSUM test on OLS residuals is formally a test for regression parameter stability. In the absence of a structural break, regression parameters should be stable over time, and residuals should exhibit the properties of white noise: mean zero, constant variance, drift around zero. The test works by computing the cumulative sum of deviations of the residuals from a mean zero and comparing that sum to a predefined limit. If deviations are systematically positive or negative, a break occurred. I use a time-trend regression: Rev(t) = B0 + B1*t + e(t). Unlike a Chow or QLR test, the CUSUM test doesn't identify where the break occurs, only that it has occurred.

FX, Global Rates, and Other, the fitted SARIMAX and trend-seasonality regression models should identify statistically significant seasonality. I plot revenues, in levels, of desks identified as non-stationary:

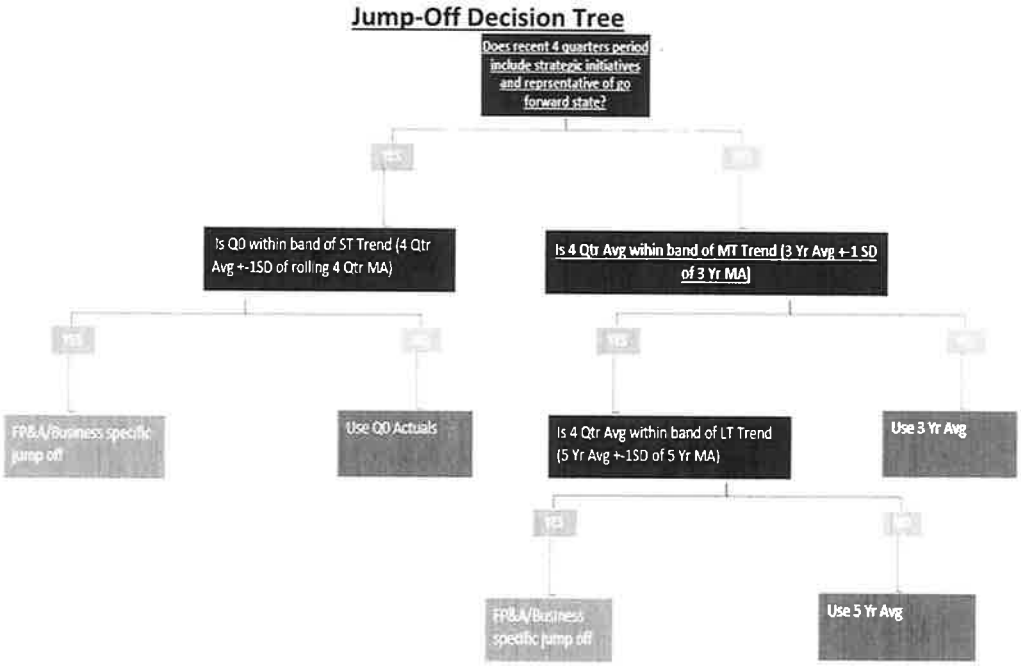


Several desks, most notably Equity Admin and Equity Derivatives, show time-varying volatility. I repeat the tests on first-differenced data[3]. Seasonality is present for all desks except Equity Admin, F&S, G10 MT, and GSP Other. First-differenced revenues are stationary for all but Equity Cash, and there are no structural breaks for any desk. The Hamilton filter requires a series that is I(4), and all fourth-differenced desk revenues are stationary.

[3] First-differencing doesn't induce seasonality, but it may remove linear trends that mask seasonality. See the Appendix for complete results.

Current Framework Deficiencies

PPNR markets currently uses a decision tree framework to determine jump-off revenue:



Decision tree approved by MRM for PPNR markets model jump off determination.

Despite the structure of the tree, this approach collapses to trailing averages of actual revenues at different horizons. A '4QMA' is a linear filter that convolves observed revenue with fixed coefficients:

$$\widehat{Q}_t = \sum_{j=-\infty}^{j=\infty} \beta_j \mathrm{Rev}_{t-j}$$

$$\beta_j = 0.25 \cdot \mathbf{1}_{j=0:3}$$

There are several deficiencies. First, the choices of βs, standard deviation bands, and windows are arbitrary. Second, the assessment of whether Q_0 is representative of a go-forward state – the top node of the tree - is entirely subjective. Third, averaging presumes the time series of actuals is ergodic, meaning that sample moments can be computed from a single realization of the series. Minimally, such an assumption is unwarranted. It is likely false because revenue actuals in levels are non-stationary for several desks, so there are no time-invariant population moments. Fourth, it permits 68% of the actuals' cross-sectional distribution as a jump-off. Floors and caps in the revised framework beg the question of how actuals should be bounded. Fifth, the tree offers no quantitative prescription in 40% of cases.

Advantages include MRM approval, interpretability, and ease of use. The structure of the decision tree is conducive to programmatic[4] implementation. I select the 'Yes' branch at the first level of the tree, evaluate the tree for each desk using historical actuals, and track the outcomes. Outcomes are as follows for the case where the recent 4Q actuals are representative of the go-forward state:

[4] See the Appendix for sample output for the desks using the accepted framework, in which Q_0's representativeness of the business's go forward state is set randomly. The Appendix also contains a proposed revised decision tree.

Decision Tree Outcomes, Recent 4Q Representative of Go Forward State

Level 1 "Yes" Branch Outcomes		
FP&A/Bus. specific JO	14/17	Commodities, DM - Credit Trading, DM - Structured Products Trading, EMCT, Equity Admin, F&S, FDC – Clearing, FX, G10 MT, GSP Other, Global Rates, LMT, Other, Prime Finance
Q_0 Actuals	3/17	Equity Cash, Equity Derivatives, Finance Desk
3Y Avg.	-	-
5Y Avg.	-	-

Summary of decision tree outcomes for market desks, using historical actuals and "Yes" at first branch.

For 14/17 desks, the tree offers no prescribed jump-off. When the recent 4Q is not representative, the tree prescribes the 3-year average, 5-year average, and a business jump-off in almost equal proportion.

Decision Tree Outcomes, Recent 4Q Not Representative of Go Forward State

Level 1 "No" Branch Outcomes		
FP&A/Bus. specific JO	6/17	Commodities, EMCT, FDC – Clearing, GSP Other, Global Rates, Prime Finance
Q_0 Actuals	-	-
3Y Avg.	5/17	DM - Credit Trading, Equity Cash, Equity Derivatives, F&S, Finance Desk
5Y Avg.	6/17	DM - Structured Products Trading, Equity Admin, FX, G10 MT, LMT, Other

Summary of decision tree outcomes for market desks, using historical actuals and "No" at first branch.

Many prescribed jump-offs when the recent 4Q is not representative are longer historical averages. If the most recent year is not relevant, then averaging over the most recent year and earlier years, which are farther from the business's current state, does little to ameliorate the disutility of the recent 4Q.

I assess the 4Q jump off approach in two further regards. First, I quantify the reduction in variance from smoothing with a 4Q average. If a 4QMA removes noise, a 4QMA time series should show considerably lower sample variance than the that of the entire unsmoothed time series. The reduction in variance is marginal for FDC – Clearing and LMT, and largest for DMCT, EMCT, and Other; intuitively, desks with considerable trading and hedging revenue should exhibit the most variance reduction.

4QMA Smoothing Variance Reduction

Desk	$\hat{\sigma}^2_{Rev(t)}$	$\hat{\sigma}^2_{4QAvg.Rev(t)}$	Reduction	Desk	$\hat{\sigma}^2_{Rev(t)}$	$\hat{\sigma}^2_{4QAvg.Rev(t)}$	Reduction
FDC - Clearing	1,320.48	1,122.91	15%	Global Rates	229,700.55	92,664.34	60%
LMT	72,193.99	60,458.22	16%	Equity Admin	9,108.28	3,547.23	61%
Finance Desk	1,205.04	785.84	35%	DMSP	55,096.60	20,954.24	62%
Prime Finance	4,731.62	3,027.27	36%	GSP Other	9,361.50	3,426.69	63%
Commodities	13,964.44	8,915.70	36%	Equity Cash	3,695.17	1,295.28	65%
F&S	13,785.58	8,722.29	37%	DMCT	9,712.72	3,050.36	69%
G10 MT	22,267.48	11,043.81	50%	EMCT	2,033.69	587.14	71%
FX	30,434.84	14,382.99	53%	Other	19,085.67	4,799.35	75%
Equity Derivatives	45,071.19	21,018.17	53%				

Sample variance of revenue time series per desk and 4Q average of revenue time series per desk. Revenue in $MN, 2010 Q1 – 2024 Q4 (inclusive), T=60. 4Q series is length T=57. Reduction is 1 - 4Q Var./ Rev. Var. in percent. Desks sorted by least to greatest reduction.

7 out of 17 desks show a reduction in variance less than or equal to 50%. 4QMA smoothing is less effective for Finance Desk, Prime Finance, Commodities, F&S, and G10 MT than other desks.

Second, I examine the one-quarter ahead forecasting power of 4QMA smoothed revenue compared to that of lagged revenue. The hypothesis is that if 4QMA smoothing extracts a meaningful signal from a short, temporal subsample of noisy data, it should have greater predictive power than the most recent observation. The point of the jump off framework just is to avoid using one observation because it may or may not be indicative of the business's revenues going forward. For each desk, I compute one quarter ahead test MAPE and R^2 using two predictors for revenue in levels:

$$(1)\ \widehat{\mathrm{Rev}}_{t+1} = \mathrm{Rev}_t$$
$$(2)\ \widehat{\mathrm{Rev}}_{t+1} = \frac{\mathrm{Rev}_t + \mathrm{Rev}_{t-1} + \mathrm{Rev}_{t-2} + \mathrm{Rev}_{t-3}}{4}$$

As a complement, I perform one-quarter ahead regression predictability tests using the change in lagged revenue and lagged smoothed revenue. If the hypothesis is correct, predictor (2)[5] should exhibit lower MAPE and higher R^2, and the lagged smoothed revenue regression should have higher R^2:

$$(1)\ \Delta\mathrm{Rev}_t = \alpha_0 + \alpha_1 \Delta\mathrm{Rev}_{t-1} + u_t$$
$$(2)\ \Delta\mathrm{Rev}_t = \beta_0 + \beta_1 \Delta\widetilde{\mathrm{Rev}}_{t-1} + e_t$$
$$\widetilde{\mathrm{Rev}}_{t-1} \equiv \frac{\mathrm{Rev}_{t-1} + \mathrm{Rev}_{t-2} + \mathrm{Rev}_{t-3} + \mathrm{Rev}_{t-4}}{4}$$

Results are mixed. Measured by mean absolute percentage error, 4QMA revenue is a better predictor than revenue of the next quarter's revenue for 11 desks: Commodities, DMCT, EMCT, Equity Admin, Equity Cash, Equity Derivatives, FX, G10 MT, Global Rates, Other, Prime Finance, and a worse predictor for 6 desks: DM Structured Products Trading, F&S, FDC Clearing, Finance Desk, GSP Other, and LMT.

4QMA Smoothing Predictability Tests

Desk	1Q Ahead MAPE		1Q Ahead R^2		Reg. R^2	
	(1) Rev_{t-1}	(2) 4Q Rev_{t-1}	(1) Rev_{t-1}	(2) 4Q Rev_{t-1}	(1) $\Delta\mathrm{Rev}_{t-1}$	(2) Δ 4Q Rev_{t-1}
Commodities	1.65	**1.23**	0.06	0.29	**0.16**	0.15
DMCT	0.60	**0.57**	-0.49	0.03	**0.08**	0.01
DMSP	**0.62**	0.82	-0.44	0.08	**0.12**	0.11
EMCT	195.42	**111.98**	-1.08	0.15	**0.45**	0.06
Equity Admin	2.36	**1.19**	-0.12	0.25	0.06	**0.11**
Equity Cash	0.18	**0.14**	-0.25	0.11	**0.14**	0.00
Equity Derivatives	3.83	**3.22**	-0.42	0.26	**0.29**	0.12
F&S	0.14	0.14	0.41	0.35	0.08	**0.10**
FDC - Clearing	**0.11**	0.13	0.80	0.82	0.06	**0.09**
FX	0.15	**0.13**	-0.27	0.01	**0.15**	0.11
Finance Desk	**0.25**	0.27	0.35	0.14	**0.04**	0.03
G10 MT	1.40	**1.10**	-0.43	0.12	**0.48**	0.34
GSP Other	**1.24**	2.10	0.17	0.57	**0.24**	0.01
Global Rates	2.46	**1.79**	-0.66	-0.00	**0.30**	0.18
LMT	0.18	0.18	0.79	0.71	**0.11**	0.03
Other	6.12	**3.57**	-0.45	0.13	**0.12**	0.05
Prime Finance	0.15	**0.13**	0.07	0.36	0.17	**0.18**

Mean absolute percentage error and R^2 for one quarter ahead forecasts of revenue in levels using lagged revenue and lagged 4QMA revenue as predictors, and regression coefficient of determination for predictability tests of one quarter ahead change in revenue using lagged change in revenue and lagged change in 4QMA as covariates. Forecast tests use T=56 observations, 2011 Q1 – 2024 Q4, regressions use T=55 observations, 2011 Q2 – 2024 Q4 and Newey-West standard errors with 4 lags. First differencing a 4Q average is an annual difference.

[5] A lagged trailing average avoids any data leakage between test and train sample. As the time subscripts indicate, the predictor of revenue at t+1 only has exposure to revenue at t and earlier periods. For certain smoothing methods, e.g. the HP filter, this is not the case. The one step ahead predictability tests are not applicable to two-sided smoothing filters that use all data from t=1...T to smooth revenue at arbitrary t.

The desks with the largest differences in MAPE between (1) and (2) coincide with those exhibiting the largest reduction in sample variance from smoothing. 4QMA as a predictor for EMCT and Other show reduction in variance of 71% and 75%, respectively. 4Q averaging can be effective for these desks.

However, the regression tests show that the change in lagged revenue explains more of the variation of the change in revenue than the change in lagged 4QMA revenue for all desks except Equity Admin, F&S, FDC – Clearing, and Prime Finance. The gap in R^2 values is marginal for some desks. The pattern is attributable to the consequence of first-differencing a 4QMA(t) transformation, which produces $0.25(Rev_t - Rev_{t-4})$: annual change has little predictive power on contemporaneous quarterly change.

If the short run dynamics of the series captured in a 4Q average have such little predictive power, it is inadvisable to use the longer 3-year and 5-year horizon averages prescribed by the decision tree. Given that the decision tree either does not produce a jump-off value, or recommends averages lacking statistical evidence attesting to their ability to extract meaningful signals from noise or capture time series dynamics with predictive power of note, there is ample room to improve the jump-off framework.

Methods

Let Y(t) represents a noiseless component or signal, and W(t) a noise term devoid of economic content. Assuming W(t) is mean-zero, constant variance, and serially uncorrelated, an additive revenue model is:

$$\text{Revenue}(t) = Y(t) + W(t)$$

Smoothing methods[6] produce a fitted time series for revenue that eliminates noise W(t). The last fitted value, at $t=Q_0$, is the jump-off revenue used in model forecasts. The general procedure is as follows:

1. Let S(•) be a smoothing method, Y(t) a signal, and W(t) noise.
2. Compute S(Rev.(t)) to obtain $\tilde{Y}(t)$, an estimate of signal Y(t).
3. Assess the quality of $\tilde{Y}(t)$ based on its properties and those of residuals e(t) = Rev(t) - $\tilde{Y}(t)$

A broad specification covers edge cases for noise in actual Q_0. Desks generate revenue from market making, in which they maintain inventories of securities, buy from sellers and sell to buyers, profiting off the bid-ask spread. They generate fees from executing client transactions and performing services, such as margin lending or underwriting. Noise in revenue can therefore stem from (among other sources):

- Idiosyncratic Market Shocks: Unforeseen market conditions, often triggered by exogenous shocks, require desks to reposition and adjust their hedges and exposures, adversely affecting revenue. For example, the announcement of a tariff policy can prompt a sell-off of a particular currency by clients, or a central bank unexpectedly raising policy rates for a currency used in carry trades can lead clients to unwind their positions, impacting FX revenue. Idiosyncratic shocks do not exhibit persistence and don't materially impact desk revenue going forward. In contrast, Y(t) encompasses systematic market conditions pertinent to revenue generation.
- Different go-forward market conditions or business needs: A desk may exit poorly performing positions before year end. Quarter specific regulation, accounting, liquidity, capital and risk management concerns can drive position changes and therefore aberrational revenue at Q_0.
- Measurement Error: revenue figures used for modelling purposes are the product of many reporting systems overseen by different teams. Revenue figures can be revised upward or downward.

A common smoothing method for time series data is the moving average. PPNR markets models use moving average smoothed revenue as a dependent variable in regression models and to compute jump-off values for forecasts. Prior work shows the Slutzky-Yule effect can be especially pronounced for volatile revenue time series. Taking an MA can temporally displace revenue shocks, destroying co-movement with macroeconomic variables, and induce spurious oscillations. Moving average jump-off values can be inflated or deflated on account of one outlier within the 4Q window, despite Q_0 actual being perfectly reasonable. This, and other deficiencies, motivates a more sophisticated approach.

I consider a host of smoothing methods to separate Y(t) from W(t), including conventional time series methods such as SARIMAX (Seasonal Autoregressive Integrated Moving Average with eXogeneous regressors), and deterministic trend-seasonality regression, signal processing methods including Wavelet and Fast Fourier Transforms, time series filters, specifically the Hodrick-Prescott and Hamilton

[6] I use the term smoothing roughly. In the frequency domain, smoothing refers to removing higher frequencies components of a signal while retaining lower ones. The primary objective is akin to denoising: preserve signal characteristics but remove signal noise regardless of frequency.

filters, and Lowess. The MA and a cubic spline are baseline methods for comparison. I outline the properties of a levels macroeconomic variable regression which leverages the existing change model.

Summary of Methods

Method	Description	Hyperparameters
Decision Tree	Compare Q_0 to summary statistics of historical actuals.	Tree structure, historical windows length, number of standard deviations
Auto SARIMAX	Autoregressive integrated moving average model, incorporating seasonality and exogenous regressors. Use exclusively endogenous regressors for comparison purposes.	Seasonal and non-seasonal order of integration and AR/MA lag orders, seasonal lag order
Trend/Seasonality Reg.	Regress revenue on seasonality indicators, linear and quadratic time trends.	None
Levels MEV Reg.	Regress revenue on linear time trend and macroeconomic variables from forecast model.	None
Lowess	Perform a WLS regression of neighboring observations on time, assigning more weight to temporally closer observations.	Fraction k/n
Hodrick-Prescott Filter	Decompose revenue into a trend and cycle by minimizing a function of difference between revenue and trend and second-differenced trend.	Smoothing Parameter λ
Hamilton Filter	Project revenue on a constant and recent values of revenue using OLS.	*p* most recent values, forecast horizon *h*
Wavelet	Represent time series in scale and time domain, zero out noise coefficients, inverse transform.	Wavelet family, threshold level λ
Fast Fourier Transform	Represent time series in frequency domain, zero out high frequency components, and inverse transform.	Periodogram peak or threshold frequency
Moving Average	Smoothed revenue at t is the average of revenue at t, t-1, t-2, and t-k.	k lags
Cubic Spline	Smooth using piecewise continuous function constituted by cubic polynomials.	λ, a non-negative smoothing parameter

Filter (HP, Hamilton), regression (MEV levels, trend seasonality, Lowess), time series (SARIMAX), and signal processing (Wavelet, FFT) methods. Hyperparameters are modeler determined configurations for the model's behavior and structure. Parameters are determined by the model.

Although the SARIMA[7] and trend seasonality methods are domain agnostic, the Hamilton and Hodrick-Prescott filters are specifically designed for macroeconomic and business cycle time series. Wavelets and FFT are not conventionally applied to financial time series. Most methods have hyperparameters to tune, and the process for doing so can add complexity, model risk, and lessen ease of use and interpretability. For each method, I provide mathematical exposition and implementation details, the hyperparameter tuning procedure where appropriate, and comment on method applicability.

[7] I don't use exogenous regressors in fitted SARIMAX models. Exposure to macroeconomic covariates can only advantage the SARIMAX relative to other methods that access only revenue at leads and lags and time trends. Consequently, I use SARIMA interchangeably going forward.

Auto SARIMAX

A natural model is a Seasonal Autoregressive Integrated Moving Average with eXogenous regressors. A SARIMAX is a standard time series model that incorporates multiplicative seasonality and exogenous regressors. The PMDArima library will automatically fit a SARIMAX model to revenue actuals in levels, algorithmically performing many steps[8] a time series modeler would. In the notation below and in estimation, I do not incorporate exogenous regressors. The complete model specification is SARIMA(p, d, q)x(P, D, Q)$_s$, for seasonal autoregressive and moving average lags P and Q, seasonal integration order D, non-seasonal AR and MA lags p and q, non-seasonal order of integration d, and seasonal lag order s.

$$\Phi(B^s)\phi(B)\nabla_s^D\nabla^d Y_t = \delta + \Theta(B^s)\theta(B)\epsilon_t$$
$$\epsilon_t \sim \mathrm{WN}(0, \sigma^2)$$
$$\nabla_s^D = (1 - B^s)^D$$
$$\Phi(z) = 1 - \Phi_1 z - \ldots - \Phi_p z^p$$
$$\Theta(z) = 1 + \Theta_1 z + \ldots + \Theta_Q z^Q$$

The SARIMA has several advantages. First, it explicitly models white noise. It can be further represented as a structural model if there is additional error (such as measurement error). Second, the form is flexible, spanning an AR(1) to arbitrarily complicated time series models. Third, the calibration process is quick and entirely automated[9]. The disadvantages are the reliance on endogenous regressors and use of multiplicative seasonality, which assumes the seasonal effect scales with the level of the series. Although PPNR modelers are averse to endogenous regressors for forecasting out of sample, there is lesser concern when their application is limited to an in-sample fit. To compare SARIMA results to those of other methods, I use only endogenous regressors. After manually assessing each desk's revenues time series properties[10], I apply the auto SARIMA method to revenue actuals without modification.

Auto SARIMA Model Fits per Desk

Desk	Order	Seasonal Order	Intercept Included	Seasonality Present	Serial Correlation	Stationary	Structural Break Present
Commodities	(0, 1, 1)	(0, 0, 0, 4)	False	True	True	True	False
DMCT	(2, 1, 0)	(1, 0, 1, 4)	True	True	True	True	False
DM Struct. Prod. Trading	(3, 1, 0)	(0, 0, 1, 4)	True	True	True	False	False
EMCT	(1, 1, 1)	(1, 0, 1, 4)	True	True	True	False	False
Equity Admin	(3, 1, 0)	(2, 0, 0, 4)	False	False	True	True	False
Equity Cash	(3, 2, 1)	(0, 0, 1, 4)	False	True	True	False	False
Equity Derivatives	(3, 1, 0)	(0, 0, 1, 4)	True	True	True	False	False
F&S	(1, 1, 1)	(0, 0, 0, 4)	False	False	True	False	False

[8] Steps include testing for seasonality, identifying the seasonal lag order s and differencing, if appropriate, testing for stationarity and identifying the non-seasonal order of integration, differencing again if appropriate, performing a grid search using information criterion to select seasonal and non-seasonal lag order, estimating parameters, and examining diagnostics measures such as residual serial correlation.

[9] I use the same parameters for each auto_arima() function call for all desks. Starting p = 0, select d based on ADF test, starting q = 0, maximum p = 4, maximum d = 2, maximum q = 4, starting P = 0, select D based on OCSB test, starting Q = 0, maximum P = 2, maximum D = 1, maximum Q = 2, no maximum order, seasonal period m = 4, AIC, 5% test level, ADF test, stepwise selection, automatically select intercept.

[10] See the Appendix for test details. I use a Ljung-Box test for serial correlation, Augmented Dickey-Fuller test for a unit root, Kruskal-Wallis non-parametric test for seasonality, and a CUSUM test on the residuals of a time trend regression to test for a structural break.

FDC - Clearing	(0, 1, 1)	(0, 0, 2, 4)	True	True	True	False	False
FX	(0, 1, 1)	(0, 0, 0, 4)	False	True	True	True	False
Finance Desk	(0, 1, 0)	(0, 0, 0, 4)	False	True	True	False	True
G10 MT	(1, 1, 0)	(1, 0, 0, 4)	False	False	True	False	False
GSP Other	(1, 0, 1)	(1, 0, 0, 4)	False	False	True	True	False
Global Rates	(0, 1, 1)	(1, 0, 1, 4)	False	True	True	False	False
LMT	(1, 1, 0)	(0, 0, 0, 4)	False	True	True	False	True
Other	(0, 1, 1)	(2, 0, 0, 4)	False	True	False	True	False
Prime Finance	(0, 1, 1)	(0, 0, 0, 4)	False	True	True	True	False

Auto SARIMA order (p,d,q) and seasonal order (P,D,Q,s) per markets desk. All models fit using same parameters for function call and 2010 Q1 – 2024 Q4, inclusive (T=60). Additional columns denote the presence of an intercept in the automatically fitted model, the presence of seasonality, serial correlation, and/or a structural break, and stationarity. Auto SARIMA selects order and seasonal order of integration using statistical tests, specifically OCSB for seasonality and ADF for stationarity, and selects lag order thereafter by minimal AIC. Preliminary results showed low lag orders, hence the use of AIC over BIC. BIC penalizes model complexity; AIC prioritizes predictive power. For quarterly data, s=4.

Manual test results are sanity checks for the fitted auto SARIMA models. A time series with a structural break, for example, should be differenced, and if seasonality is present, the auto SARIMA's seasonal order should be non-zero. I note the following. First, integration and lag order are generally low. All models have seasonal integration order 0, and only equity cash is I(2). The largest model is equity cash, with an order of 7. Second, Finance Desk and LMT exhibit structural breaks. As expected, the auto SARIMA fits are I(1). Third, all fits for desks with non-stationary revenue time series are minimally I(1). Only GSP Other is I(0), and its time series is stationary in levels. Fourth, when the Kruskal-Wallis test and auto SARIMA seasonal order differ, the auto SARIMA identifies non-zero seasonal lag order that the KW test does not for 3 desks, and fails to identify seasonality where the KW test does for 5 desks. If KW results are ground truth, auto SARIMA has a higher false negative rate for detecting seasonality[11] than false positive rate. See the confusion matrix below. The case of perfect alignment would be off-diagonal counts of 0. Precision is 8/11, recall is 8/13. Auto SARIMA under diagnoses seasonality.

Auto SARIMA Seasonality Confusion Matrix

Seasonality Present (KW)	Non-Zero Seasonal Order (Auto SARIMA)	
	False	True
False	'F&S' (1/17)	'Equity Admin', 'G10 MT', 'GSP Other' (3/17)
True	'Commodities', 'FX', 'Finance Desk', 'LMT', 'Prime Finance' (5/17)	'DM - Credit Trading', 'DM - Structured Products Trading', 'EMCT', 'Equity Cash', 'Equity Derivatives', 'FDC - Clearing', 'Global Rates', 'Other' (8/17)

Confusion matrix for auto SARIMA's diagnosis of seasonality in a desk's time series, measured by the case of a non-zero seasonal order (P,D,or Q > 0), compared to the results of a Kruskal-Wallis non-parametric test for seasonality (ground truth).

Fitted models broadly accord with the results of manual time series tests. Auto SARIMA does a reasonable job of modelling the time series of revenue actuals for all desks. No single fit strains credulity, but the preponderance of autoregressive components and the magnitude of their coefficients produces in sample fits for Finance Desk and DMCT that resemble the original series shifted forward one quarter. All other in sample fits pass a visual plausibility test. Although visual plausibility is neither a necessary nor sufficient criterion for method performance, it is additional evidence in favor of a method.

[11] Auto SARIMA uses an OCSB test for seasonality. The test resembles an ADF test but for a seasonal unit root, meaning first-difference at the specified seasonal lag. See Osborn D.R., Chui A.P.L., Smith J., and Birchenhall C.R. (1988). Seasonality and the order of integration for consumption, Oxford Bulletin of Economics and Statistics 50(4):361-377.

Fast Fourier Transform

Smoothing approaches considered thus far operate in the time domain. FFT is a frequency domain approach to time series analysis, in which the regular components of the series are expressed as periodic variations of an underlying phenomenon. Frequency analysis[12] relies on the following concepts:

- Cycle: one complete period of a sine or cosine function, defined over a unit time interval
- ω: A frequency index, defined in cycles per unit time. One cycle per year, where observations are quarterly, has ω = 0.25 cycles per observation, one cycle every 12 months corresponds to ω = 0.083. Frequency is often expressed in Hz, which is one cycle per second.
- Period: the number of points in a cycle, or 1/ ω, e.g. 4 quarters per cycle, or 12 months per cycle.
- A: amplitude or height
- $\boldsymbol{\Phi}$: the phase which determines the starting point of the cosine function
- Folding frequency: the highest frequency observable in discrete sampling. Determining a cycle requires at least 2 points, so the folding frequency is 0.5 cycles per point.

Let $x_1 \ldots x_n$ be a discrete time series sample. For n odd, and t=1...n, it is exactly the case that:

$$x_t = a_0 + \sum_{j=1}^{(n-1)/2} [a_j \cos(2\pi t j/n) + b_j \sin(2\pi t j/n)]$$

and for n even, that

$$x_t = a_0 + \sum_{j=1}^{(n/2-1)} [a_j \cos(2\pi t j/n) + b_j \sin(2\pi t j/n)] + a_{n/2}(-1)^t$$

where the coefficients are as follows:

$$a_0 = \bar{x}$$

$$a_j = \frac{2}{n} \sum_{t=1}^{n} x_t \cos(2\pi t j/n)$$

$$b_j = \frac{2}{n} \sum_{t=1}^{n} x_t \sin(2\pi t j/n)$$

The frequencies $\omega_j = j/n$ are fundamental frequencies. Coefficients are defined at each fundamental frequency. The scaled periodogram[13] measures the magnitude of the frequency components of the periodic representation of the sample:

$$P(j/n) = a_j^2 + b_j^2$$

For purpose of smoothing, a periodogram's utility lies in indicating the dominant frequencies of the sample, evident from large values of the periodogram, and the smaller values associated possibly with

[12] FFTs answer the question of what frequency explain a majority of temporal variation in observed data. I provide an illustrative example using daily temperature data. The data will show an annual cycle (a cycle that occurs once a year): temperature is lowest in winter, rises through the spring, peaks in the summer, and declines in the fall. There may be higher frequency cycles, e.g. a sub-annual cycle that occurs twice a year, or lower frequency cycle, e.g. a weather cycle that occurs once every 4 years. To quantify variation in the data associated with cycles at different frequencies, one can regress daily temperature on sine and cosine terms with periods corresponding to the frequency of interest. A twice a year cycle for an annual time scale measured in days (365) has a period of 182 days. For n time series observations, there are n/2 candidate sine and cosine regressor terms. Examining higher frequency variation in temperature would require a higher sampling rate; intra-day temperature variation requires intra-day temperature measurements, and so forth.

[13] The theoretical rationale is beyond the scope of this exercise. It suffices to consider the periodogram for the purposes of analyzing the sample time series without consideration of the spectral density. Consider the periodogram as the sample spectral density of x(t), where the spectral density is a unique, monotonically increasing function that characterizes the stationary time series' population autocovariance in the frequency domain. Exposition presented here mostly follows Shumway & Stoffer 4th Edition, see Appendix C.

noise. In practice, it is quickest to use a Discrete Fourier Transform to obtain the frequency representation. Given data $x_1 \ldots x_n$, define the DFT as:

$$d(\omega_j) = n^{-1/2} \sum_{t=1}^{n} x_t e^{-2\pi i \omega_j t}$$

for j = 0, 1, ..., n-1. The DFT of a discrete sample is a sequence of complex values, of the same length as the input sample. The DFT is a linear transformation that provides a 1:1 mapping from the time domain to the frequency domain. Correspondingly, the inverse DFT is:

$$x_t = n^{-1/2} \sum_{j=0}^{n-1} d(\omega_j) e^{2\pi i \omega_j t}$$

for t = 1...n. The periodogram is the squared modulus of the DFT:

$$I(\omega_j) = |d(\omega_j)|^2$$

and the scaled periodogram is the periodogram multiplied by (4/n). The periodogram can be expressed as the sum of real and imaginary components using Euler's identity:

$$d_c(\omega_j) \equiv n^{-1/2} \sum_{t=1}^{n} x_t \cos(2\pi\omega_j t)$$

$$d_s(\omega_j) \equiv n^{-1/2} \sum_{t=1}^{n} x_t \sin(2\pi\omega_j t)$$

$$d(\omega_j) = d_c(\omega_j) - i d_s(\omega_j)$$

$$I(\omega_j) = d_c^2(\omega_j) + d_s^2(\omega_j)$$

where the coefficients follow:

$$a_j = \frac{2}{\sqrt{n}} d_c(\omega_j)$$

$$b_j = \frac{2}{\sqrt{n}} d_s(\omega_j)$$

Despite complicated mathematical underpinnings, the denoising algorithm is quite simple:

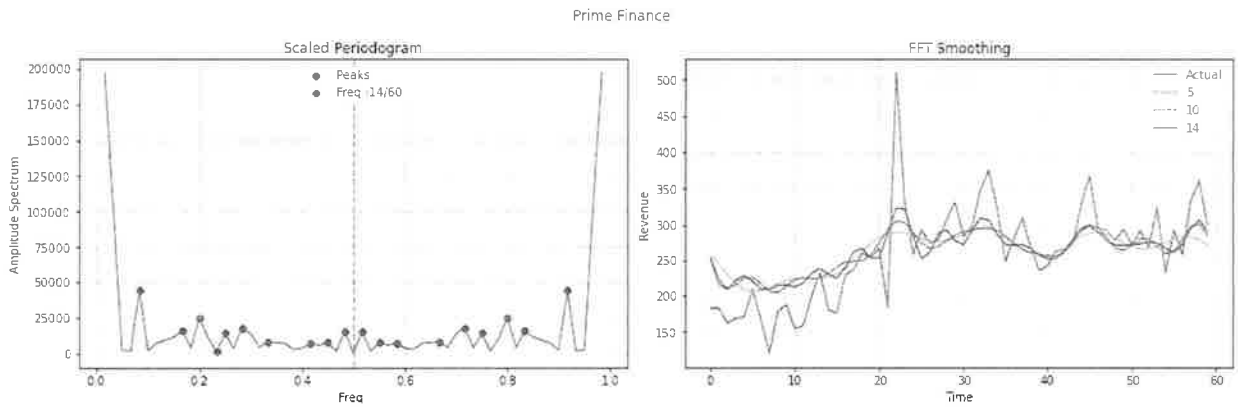
1) Detrend the time series of actuals. Regress revenue on an intercept and linear time trend: $Rev_t = B_0 + B_1 t + e_t$, detrended actuals are $Rev_t - \hat{B}_0 - \hat{B}_1 t$. Doing so eliminates induced extremely low frequency components.
2) Use the DFT to obtain a frequency domain representation of the time series of revenue actuals. For input actuals of length n, the DFT is a length n sequence of complex values.
3) Identify a dominant fundamental frequency at 0 .... n-1, call it the cutoff frequency c, after which frequencies are associated with noise. To identify the cutoff frequency, use:
    a. Examine the periodogram and select the frequency associated with a peak
    b. Select the first frequency at which the periodogram's value falls below some fraction of the average periodogram value (mean power)
    c. Cross-validation
4) Apply a rectangular low-pass filter[14] to the DFT, effectively zeroing out DFT values at frequencies beyond the cutoff frequency.

[14] A low pass filter lets low frequencies pass through. A rectangular or boxcar filter zeros out observations outside of the filter's window. Alternative filters, such as a Gaussian filter, weight mostly heavily observations closest to the cutoff. Preliminary testing shows no meaningful performance benefit from more complicated filters.

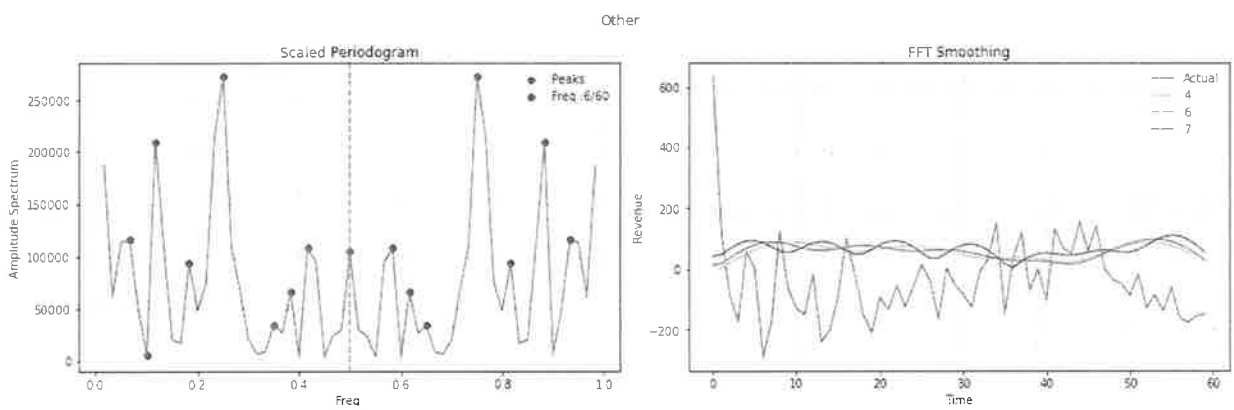
$$
\begin{aligned}
x_{t:1\ldots n} &= (x_1, \ldots, x_n) \\
x_{j:0\ldots n-1} &= (d(\omega_0), \ldots, d(\omega_{n-1})) \\
\tilde{x}_{j:0\ldots n-1} &= H \cdot x_{\text{Freq}} \\
H &= \begin{cases} 1 & j \le c, j \ge (n-1) - c \\ 0 & j > c, j < (n-1) - c \end{cases}
\end{aligned}
$$

5) Perform an inverse DFT to map the modified frequency domain representation back to the time domain. Add in the linear trend and the resulting time series is the smoothed revenue actuals.

The key issue is how to select the cutoff frequency. It may be the case that the dominant frequency in the signal is low, in which case using the graphical approach of the first or second peak of the periodogram is convenient. However, subsequent frequencies may not be noise. Consider periodograms for the Other and Prime Finance revenue series. Prime Finance's smoothed time series, using the periodogram approach, is reasonable because the periodogram shows low power after its first peak.



Prime Finance, Q1 2010 – Q4 2024 inclusive (T=60) observations. Left: Periodogram, frequency scale is 1/60...59/60. Red dots indicate local maxima. Green dot is first frequency at which periodogram falls below 10% of the mean power. Right: smoothed revenue in time domain, using cutoff frequencies associated with 1st and 2nd peak of periodogram and mean power approach. Legend reports cutoff frequencies.



Other, Q1 2010 – Q4 2024 inclusive (T=60) observations. Left: Periodogram, frequency scale is 1/60...59/60. Red dots indicate local maxima. Green dot is first frequency at which periodogram falls below 10% of the mean power. Right: smoothed revenue in time domain, using cutoff frequencies associated with 1st and 2nd peak of periodogram and mean power approach. Legend reports cutoff frequencies.

Other's denoised actuals are a poor fit because the periodogram has higher power after its first peak. Selecting a lower cutoff frequency results in more smoothing, a higher cutoff frequency, less smoothing. Using n-1 as the cutoff frequency results in no smoothing. Methods 3(a) and 3(b) require considerable

intervention (e.g. manual examination of each desk's periodogram[15], selecting a fraction of mean power) and invites subjective judgements, contradicting the first guiding principle.

I devise a cross-validation procedure[16] to automatically select c. The procedure iteratively removes an actual observation, replaces it with an interpolation, smooths, and compares smoothed to actual to assess the effect of the cutoff frequency on the transformation's ability to reconstruct the underlying signal. If the interpolation for the omitted observation produces a distant stand in, the cutoff frequency should be set such that the deleted observation can be reconstructed from the transformed signal. A deficit of this procedure is that for certain desks where the signal is quite smooth, or high frequency components are captured by surrounding immediate observations, the procedure produces an optimal cutoff equal to the folding frequency, resulting in a smoothed series identical to actuals. I include a penalty weight to select lower cutoffs. The table below reports candidate cutoffs per desk:

Candidate FFT Smoothing Cutoff Frequencies Per Desk

Desk	Periodogram Peak 1 Freq.	Periodogram Peak 2	Mean Power Freq.	CV Optimal Freq.
Commodities	6/60	9/60	4/60	**2/60**
DM - Credit Trading	2/60	6/60	23/60	**1/60**
DM SP	4/60	8/60	9/60	**2/60**
EMCT	5/60	7/60	8/60	**1/60**
Equity Admin	**4/60**	9/60	7/60	**4/60**
Equity Cash	**4/60**	6/60	18/60	**4/60**
Equity Derivatives	4/60	8/60	19/60	**2/60**
F&S	**4/60**	6/60	18/60	9/60
FDC - Clearing	**6/60**	12/60	8/60	7/60
FX	**3/60**	6/60	9/60	6/60
Finance Desk	**4/60**	6/60	10/60	7/60
G10 MT	**4/60**	9/60	6/60	**4/60**
GSP Other	**1/60**	8/60	24/60	14/60
Global Rates	4/60	7/60	3/60	**1/60**
LMT	**3/60**	9/60	6/60	5/60
Other	4/60	7/60	6/60	**1/60**
Prime Finance	5/60	10/60	14/60	**2/60**

Cutoff frequencies suggested by graphical periodogram, mean periodogram power, and cross-validation procedures for FFT smoothing, per desk. Each desk's revenue is quarterly, 2010 Q1 – 2024 Q4 (T=60), inclusive. Candidate frequencies are j/n, for j=0...n-1, n=60. Lower frequencies produce more smoothing, higher frequencies, less smoothing. Columns 1 and 2 show the frequencies associated with the first two peaks of the periodogram. Column 3 shows the first frequency at which the periodogram's power falls below the mean power (average of periodogram values at frequencies 1 to n/2). Column 4 shows the optimal frequency prescribed by the CV procedure. Lowest in bold.

The mean power approach tends to prescribe high cutoff frequencies, and the periodogram and CV procedures, low cutoffs. For Equity Admin, Equity Cash, and G10 MT, CV and the 1st periodogram peak prescribe the same cutoff. I test two approaches to FFT smoothing, one using the first peak of the periodogram and the other using CV. Only GSP Other's CV prescribed optimal frequency is the maximum in the candidate range. GSP Other's time series of revenue shows a downward trend and no discernible pattern, consistent with hedging revenue. Most other desks only need low frequencies to meaningfully reconstruct the input signal, indicating FFT smoothing is a plausible approach for revenue actuals.

[15] A cursory glance suggests the following n^{th} periodogram peaks capture the dominant frequencies: Commodities: 4, DM – Credit Trading: 4, DM – Structured Products Trading: 4, EMCT : 5, Equity Admin: 3, Equity Cash: 5, Equity Derivatives: 5, F&S: 3, FDC – Clearing: 1, FX : 5, Finance Desk: 2, G10 MT: 1, GSP Other : 4, Global Rates: 3, LMT : 2, Other : 4, Prime Finance: 3. All periodograms can be found in the Appendix.

[16] Iterate through each time series, removing one observation at a time and replacing it with a linearly interpolated value. Smooth the series and compare the smoothed value to the actual at the omitted index. Compute the MSE. Repeat for each candidate cutoff frequency, select the CV error minimizing frequency. See the Appendix for complete details.

Wavelet

Wavelet analysis, like FFT, is an approach from signal processing. Wavelets are used to analyze, compress, and de-noise signals, with applications to images, audio, seismic, and time series data. FFT expresses a signal in the frequency domain, but gives no indication of *when* the frequencies occur, whereas wavelets represent a signal in the time and scale domain. A Fourier transform decomposes a signal into sines and cosines, which are global oscillations, and is suboptimal for analysis of the local properties of a signal, such as edges or transients. Desk revenues exhibit the sharp fluctuations and spikes that Wavelets are better suited than Fourier methods for analyzing. Wavelets are localized waves. Specifically, a wavelet is a real-valued function ψ that satisfies two mathematical criteria:

$$\int_{-\infty}^{\infty} \psi(u)du = 0$$

$$\int_{-\infty}^{\infty} \frac{|\hat{\psi}(\omega)|^2}{\omega} d\omega < \infty$$

Criteria 1 states the average of the function over the real line is 0. Intuitively, the wavelet is an oscillatory, decaying, wave. Condition 2, admissibility, is imposed on the Fourier transform of the wavelet function and ensures a signal can be analyzed and reconstructed without loss of information. The function can be relocated (translated) around t and stretched or shrunk (dilated) by τ, so the following family of functions stems from the mother wavelet ψ.

$$\psi_{\tau,t}(u) \equiv \frac{1}{\sqrt{\tau}}\psi(\frac{u-t}{\tau})$$

Define the continuous wavelet transformation of time series x(t), a real-valued function of time t, as:

$$W(\tau,t) = \int_{\infty}^{\infty} x(u)\phi_{\tau,t}(u)du$$

and the inverse transformation to reconstruct the signal in the time domain as:

$$x(t) = \int_{0}^{\infty} \left[\frac{1}{C_T^2} \int_{-\infty}^{\infty} W(\tau,u) \frac{1}{\sqrt{\tau}} \psi\left(\frac{t-u}{\tau}\right) du \right] d\tau$$

for C a constant specific to the wavelet ψ. The CWT produces coefficients at each point in the time-scale plane because the relocation and scale parameters are continuous. The Wavelet coefficient W(τ, t) is the product of the wavelet function at a specific time and scale, where scale is the width of the interval for which t is the midpoint. The structure of the CWT permits analysis of large and small features of the signal at different times. The Haar wavelet has a convenient physical interpretation as the difference between τ weighted averages of x(t) before and after t. In the case of discrete x, with T samples, the scaling parameter is restricted to powers of 2 and time restricted to integers 0 to T-1. For integers m and n, let the scaling and translation parameters be, respectively:

$$\tau = 2^m$$
$$t = n2^m$$

The discrete wavelet transformation reparametrizes the Wavelet function in terms of the integers m, n:

$$W(m,n) = \int_{\infty}^{\infty} x(u)\psi_{m,n}(u)du$$

$$\psi_{m,n}(u) \equiv 2^{-m/2}\psi(\frac{u - n2^m}{2^m})$$

and Wavelet coefficients proportional to CWT coefficients from which the discrete sample x can be recovered perfectly. The inverse transformation follows:

$$x(t) = \sum_{m,n} \psi_{m,n}(u)W(m,n)$$

To separate signal from noise, I use hard thresholding (Rangarajan et al 2002), which assumes large wavelet coefficients are meaningful signal features and small coefficients are noise. For specified threshold λ, and Wavelet coefficient d, the hard threshold operator zeros out coefficients with an absolute value below the threshold:

$$D^H(d|\lambda) = \begin{cases} 0 \text{ for } |d| \leq \lambda \\ d \text{ for } |d| > \lambda \end{cases}$$

Wavelet denoising requires a specific choice of wavelet function, the number of decomposition levels to parse noise from signal, and the hard thresholding level. I use a multilevel DWT, db4[17] wavelet, and hard thresholding with 1 level,[18] where I selected the level after trial and error. The maximum possible Wavelet decomposition level, the last level where at least one Wavelet coefficient is correct, is $\log_2(T)$, for a signal of length T. The maximum effective level used in the pywavelet DWT is $\log_2(T/(\text{db4 Length} - 1)$, where the denominator is the length of the db4 filter. For T=60 quarterly observations and the db4 function, the maximum level is 3. There exist more sophisticated denoising techniques using Wavelets, and more rigorous approaches to selecting the optimal level with hard thresholding, but the decision comes to choosing between 0 (no smoothing), 3 (excessive smoothing), 1, or 2. Across desks, I found 1 performed better than 2.

Sample size is an acute challenge with Wavelet analysis. 64 quarterly observations permit at most 6 levels, which is not enough granularity to parse signal from noise in the same manner that cutoff frequencies for FFT permits. The limitation for Wavelet denoising on markets revenue is that high frequency noise, which has a quarterly or sub-quarterly component, needs more data sampled at a higher frequency. Wavelets are a more promising candidate for monthly and daily revenue data.

[17] Daubechies-p wavelets represent polynomials of degree p-1. A db4 wavelet covers cubic polynomials, adequately capturing non-linear trends in revenue actuals. Higher p requires fewer scales and coefficients. See the Appendix 'Wavelet Intuition' for an example.

[18] Per the documentation, "The rational for the choice of levels is the maximum level where at least one coefficient in the output is uncorrupted by edge effects caused by signal extension. Put another way, decomposition stops when the signal becomes shorter than the FIR filter length for a given wavelet". See PyWavelets Documentation, "Discrete Wavelet Transform" (2025).

Trend Seasonality Regression

Additive seasonality, of the form Rev(t) = Trend(t) + Seasonality(t) + Noise(t) can be modelled with a regression of revenue on period indicator variables, a linear trend, and a quadratic trend:

$$\text{Rev}(t) = B1 \cdot I(Q1) + B2 \cdot I(Q2) + B3 \cdot I(Q3) + B4 \cdot I(Q4) + B5 \cdot t + B6 \cdot t^2 + e(t)$$

The objective for this regression is quality of in sample fit, even at the expense of model variance, hence I include a quadratic trend for to capture any non-linearity. The dummy variables are 1 if t falls into the corresponding quarter and 0 otherwise, and their coefficients can be interpreted as quarter specific intercepts, or equivalently, the average revenue in a quarter after removing the effect of the linear and non-linear trend. There is no intercept as it would be collinear with the quarterly dummies.

Advantages of this approach are as follows. First, it is easily interpreted. The conveniences of a linear regression model carry over, including measures of statistical significance and goodness of fit[19]. Second, the model is macroeconomic variable agnostic. Third, there is no calibration or fine-tuning of hyperparameters involved. Fourth, the model is easy to use and can be implemented in Excel. The disadvantages are that it provides a poor fit for certain desks, such as those with multiplicative seasonality or no discernible time trends, and when the estimated trend coefficients are positive, the models in sample fit at Q_0 suggests an upward trend that may not necessarily be representative of the business's go-forward state. I present regression output per desk.

Trend/Seasonality Regression Output (1/3)

Desk	Commodities	DMCT	DM SP	EMCT	Equity Admin	Equity Cash	Equity Derivatives
Q1	43.2641 (33.1498)	**185.6065***** (46.4687)	**691.6975***** (86.3734)	**150.7215***** (14.0879)	14.1986 (70.4636)	**381.2112***** (17.0885)	**347.0341***** (52.3905)
Q2	22.9563 (28.0373)	**100.9149***** (34.3893)	**645.3835***** (68.0826)	**102.5162***** (12.9789)	9.5040 (50.6661)	**340.5281***** (15.0134)	**199.3719***** (60.1903)
Q3	33.8556 (30.5084)	66.5317 (46.8095)	**535.6421***** (83.0156)	**111.1298***** (14.7662)	29.9831 (65.4155)	**328.2013***** (15.5952)	**184.4261***** (68.3877)
Q4	-19.4428 (28.9270)	20.6619 (46.0185)	**402.6877***** (84.1025)	**65.3285***** (13.4732)	-18.2919 (53.9952)	**337.8799***** (28.4666)	14.5562 (73.0822)
T	**5.5025**** (2.4434)	5.2999* (3.0034)	-6.7780 (4.8912)	-0.0292 (0.9119)	0.1635 (3.4679)	**-6.3552***** (1.8328)	-0.7822 (4.0440)
T^2	-0.0133 (0.0426)	-0.0828* (0.0424)	0.0139 (0.0707)	-0.0168 (0.0150)	-0.0479 (0.0468)	**0.0853**** (0.0365)	**0.1261**** (0.0626)
R^2	50.80%	43.13%	45.08%	66.91%	31.37%	37.84%	62.27%
R^2 Adj.	46.24%	37.87%	39.99%	63.85%	25.01%	32.08%	58.77%

Regression output for market desks revenue trend-seasonality model: Rev(t) = B1*I(Q1) + ... + B4*I(Q4) + B5*t + B6*t^2 + e(t). Dependent variable is quarterly revenue in levels, non-seasonally adjusted, for Q1 2010 – Q4 2024, inclusive (T=60). Intercept omitted to avoid dummy trap. Newey-West standard errors in parenthesis, * p<.1, ** p<.05, ***p<.01, with a maximum of 4 lags. Lag order selected using AIC.

Equity cash shows a statistically non-zero and negative linear trend. Non-linearity is significant at 10% in DMCT, and at 5% for equity cash, and equity derivatives. The effect is small in magnitude, although for equity derivatives it's roughly 1/6th of the linear trend. All quarterly seasonality dummies are significant for DM Structured Products trading, EMCT, and Equity Cash. Revenues are on average $40-50mm higher in Q1 than in Q2. Adjusted R^2 measures range from 25% to 64%. For the second set of desks below, all but global rates have statistically significant quarterly dummy coefficients at a 5% level. Net of the trend in revenues, the average revenue differs per quarter for most desks. Revenues are highest in Q1 except

[19] R-squared values facilitate relative quality of fit, meaning for which desks is the regression model better suited. They aren't useful for general assessment of smoothed relative to actual, as a smoothing method can trivially maximize R-squared by returning an unsmoothed series.

for F&S (Q4) and G10 MT (Q3). FX and GSP Other have large negative linear trend coefficients. Adjusted R^2 values are comparable to the first set of desks. See the Appendix for LMT, Other, and PF results.

Trend/Seasonality Regression Output (2/3)

Desk	F&S	FDC - Clearing	FX	Finance Desk	G10 MT	GSP Other	Global Rates
Q1	**442.4011*** ** (31.1113)	**37.0366*** ** (6.4770)	**1343.4107*** ** (68.5816)	**58.5299*** ** (11.0719)	**263.9227*** ** (51.3584)	**192.9873** ** (75.8475)	311.5661 (227.9598)
Q2	**457.9348*** ** (30.9131)	**29.9975*** ** (6.8083)	**1218.6158*** ** (67.1369)	**43.6513*** ** (11.7719)	**267.1754*** ** (49.2167)	**153.2410*** ** (55.9261)	183.9583 (193.9162)
Q3	**463.1557*** ** (42.3376)	**25.3473*** ** (7.2651)	**1212.0779*** ** (58.3364)	**43.2205*** ** (12.5458)	**294.6106*** ** (51.1297)	**149.8053*** ** (54.2244)	264.1178 (311.6675)
Q4	**467.6366*** ** (44.7347)	**23.1349*** ** (7.1334)	**1149.9650*** ** (66.1551)	**44.5736*** ** (12.3960)	**269.8906*** ** (41.3769)	**158.0054*** ** (57.0939)	-167.9514 (240.0670)
T	-1.0328 (2.5770)	**1.7148*** ** (0.5002)	**-13.8140*** ** (4.0011)	**4.6407*** ** (0.9431)	**-13.0382*** ** (3.5349)	**-11.3975*** ** (3.8825)	-7.2534 (13.0882)
T^2	**0.0927** ** (0.0385)	0.0037 (0.0076)	**0.1874*** ** (0.0670)	**-0.0699*** ** (0.0143)	**0.2519*** ** (0.0621)	**0.1288** ** (0.0539)	0.3298* (0.1683)
R^2	0.5322	0.8725	0.3178	0.3629	0.2903	0.5811	0.3913
R^2 Adj.	0.4888	0.8607	0.2547	0.3039	0.2246	0.5423	0.3349

Regression output for market desks revenue trend-seasonality model: Rev(t) = B1*I(Q1) + ... + B4*I(Q4) + B5*t + B6*t^2 + e(t). Dependent variable is quarterly revenue in levels, non-seasonally adjusted, for Q1 2010 – Q4 2024, inclusive (T=60). Intercept omitted to avoid dummy trap. Newey-West standard errors in parenthesis, * p<.1, ** p<.05, ***p<.01, with a maximum of 4 lags. Lag order selected using AIC.

Goodness of fit and statistical significance measures confirm the trend/seasonality regression approach is feasible as a standalone model for the jump-off. The signal Y(t) is the linear combination of indicators and trends, and its estimate is the regression's in sample fit. The weak exogeneity assumption underpinning OLS aligns with the characterization of noise W(t). Ultimately, the regression output helps inform the plausibility of model derived jump off. A model's jump-off higher than Q1 revenue for a desk, where the Q1 coefficient is noticeably larger than those of Q2 – Q4, for example, is at odds with the history of revenue actuals. Similarly, a derived jump off that is much higher than Q_{-3}, Q_{-2}, Q_{-1}, and Q_0 is more persuasive if the desk has positive estimates for the coefficients on linear and quadratic trends.

Lowess

Local weighted scatterplot smoothing is a form of nearest neighbors regression. Smoothed values are calculated using weighted least squares regression on a subset of the data local to the observation being smoothed, where temporally closer data gets more weight and farther data gets less weight. Lowess performs one regression per observation. The key hyperparameter is the fraction of nearest neighbors included in the regression. The larger the fraction, the smoother the fit; as the fraction approaches 1, the smoothed fit approaches a regression-esque linear best fit, and as the fraction approaches 0, the smoothed fit approaches the unsmoothed actuals. For data (y_t, x_t), and at each t = 1 ... n observation:

$$\min_{\beta_t^0, \beta_t^1} \sum_{j=t-\frac{k}{n}}^{j=t+\frac{k}{n}} w_j (y_j - \beta_t^0 - \beta_t^1 x_j)^2$$

$$\hat{y}_t = \hat{\beta}_0^t + \hat{\beta}_1^t x_t$$

The covariate x_t is a time index. The WLS regression is fit on observations within the region (t – k/n, t + k/n), but evaluated at only x_t. The term fraction, referring to k/n, and the span, k, are conceptually interchangeable for fixed n. The statsmodels library's implementation of Lowess smoothing by default uses tricube weighting and additional local regressions using residuals from 3 prior iterations to adjust weights. Where d is the distance between the kth nearest x_j to x_t, x_j's weight w_j is as follows:

$$w_j = \left[1 - \left(\frac{|x_j - x_t|}{d}\right)^3\right]^3$$

Advantages are as follows. First, Lowess is a commonly used smoothing method for cross-sectional data, with some precedent for use in time series data. By drawing on local observations, it is better suited to non-stationary time series data that exhibits time-varying moments than methods that use the entire sample, such as the trend/seasonality regression. Second, it operates as a smoothing method, not a separate predictive model. The disadvantages are that it requires an additional methodology for selection of the fraction, justification for choice of weighting function, and when the fraction is low, the jump off does not meaningfully differ from actual Q_0 because Q_0 is the endpoint of the sample. There are no local neighbors to the right of Q_0. To select the optimal fraction, I use cross-validation[20].

Unlike predictive models, Lowess doesn't have an immediate cross-validation procedure. To generate a smoothed value, it needs a response and covariate observation set. I use a variant of LOOCV, replacing the left-out observation with an interpolated value and comparing the Lowess smoothed value at the omitted index to the actual observation. If the fraction is wide enough to capture locally relevant properties of the neighboring observations but narrow enough to exclude temporally irrelevant observations, the two values should be close. I present optimal fractions per desk below.

Lowess CV Optimal Fraction Per Desk

0.00 – 0.10	Other 0.0 FDC – Clearing 0.07 Finance Desk 0.07 FX 0.09
0.11 – 0.20	LMT 0.11

[20] For each candidate span k = 1 ... n-1, order the sample of revenue actuals from Q(-T) to Q(0). For each observation t, omit Q(t), replace it with an interpolated value, and fit the Lowess smoother at the candidate span to get a 'predicted' smoothed value for Q(t). Compute the squared error between smoothed Q(t) and actual Q(t), and average over all t to obtain MSE per k. Select the MSE minimizing k.

	Prime Finance 0.11 F&S 0.12 GSP Other 0.16 G10 MT 0.19
0.21 – 0.30	Equity Cash 0.21 Commodities 0.29
0.31 – 0.40	Equity Admin 0.38
0.41 – 0.50	-
0.51 – 1.00	DM - Credit Trading 0.51 Equity Derivatives 0.85 Global Rates 0.9 DM - Structured Products Trading 1.0 EMCT 1.0

Optimal Lowess smoothing fraction per cross-validation procedure per desk. Fraction ranges from 0.0 to 1.0, higher means more smoothing, lower means less smoothing. Obtained using Q1 2010 – Q4 2024 revenue actuals, T = 60.

Optimal fractions for EMCT and DM Structured Products trading are 1, so the Lowess smoothed actuals for those desks' revenue resembles a straight line. For Other, the fraction is 0.0, so Lowess doesn't smooth. Equity Derivatives and Global Rates have optimal fractions above the statsmodels default of 2/3. Comparing the time series of revenue actuals to CV prescribed fractions shows a distinct pattern. Desks with more volatile and idiosyncratic revenue shocks have higher fractions (more smoothing), and desks with stable and more cyclical revenue actuals have lower fractions (less smoothing). For the latter, Lowess smoothing produces a jump off quite close to Q_0 and the method isn't especially useful.

Levels Macroeconomic Variable Regression

Given the structure of PPNR markets forecasting regression models, a natural extension to obtain a jump-off value is to undo the difference transformation. For a change model of the form:

$$\mathrm{Rev}_t - \mathrm{Rev}_{t-1} = \beta_0 + \beta_1(\mathrm{MEV}_t - \mathrm{MEV}_{t-1}) + e_t$$

The jump-off at Q_0, the number of periods elapsed since the beginning of the sample, is:

$$\widehat{\mathrm{Rev}}_{Q0} = \hat{\beta}_0 \cdot \mathrm{Q0} + \hat{\beta}_1 \mathrm{MEV}_{Q0}$$

Per the forecasting model, all variation in revenue is attributable to contemporaneous variation in macroeconomic variables and an error term. If the forecasting model is correct, the levels approach provides the only mathematically valid[21] estimate of Q_0. There is no separate model of revenue in levels.

Advantages are as follows. First, the fitted regression for Q_0 leverages the forecasting model's interpretability, exposition, and analysis. Second, no further development or validation is required. Modelers take the estimated parameters from the change model and apply them to the macroeconomic covariates in levels and a linear time trend. Third, Q_0 inherits standard OLS prediction intervals. Any adjustments to Q_0 to ensure it is representative of the go-forward state operates through the MEVs. Prediction intervals adjust with covariates. However, estimated Q_0's validity is predicated on that of the forecasting regression. Using Finance Desk and Global Rates, I demonstrate this shortcoming. For each desk, I regress first-differenced revenue on the first-differenced covariates used in current models, and compute fitted values for revenue in levels for Q4 2024 using a linear time trend and the macroeconomic variables in levels. Finance desk covariates are the federal funds rate and federal reserve 90-day AA financial commercial paper interest rate:

$$\Delta\mathrm{Revenue}_t = B_0 + B_1\Delta\mathrm{EffectiveFederalFundsRate}_t + B_2\mathrm{Fed90DayAACommericalPaperRate}_t + e_t$$

The Global Rates model uses US 3-month at 30-year expiry swaption log volatility:

$$\Delta\mathrm{Revenue}_t = B0 + B_1\Delta\mathrm{3M30YSwapVol}_t + u_t$$

First-Differenced Regression Results

Desk/Variable	Finance Desk	Global Rates
Intercept	-1.0680 (2.8958)	13.2982 (32.3882)
ΔUS_IR_FFED$_t$	-16.2441 (21.3691)	-
ΔUS_IR_3M_FCP_AA$_t$	17.8894 (14.1293)	-
ΔUS_E_3M30Y_SWP_LOG$_t$	-	**14.8974***** (3.9451)
R^2	3.82%	13.74%
R^2 Adj.	0.39%	12.23%
T	59	59

Summary of regression output for Finance Desk: ΔRev.(t) = B0 + B1*ΔUS_IR_FFED(t) + B2*ΔUS_IR_3M_FCP_AA(t) + e(t) and Global Rates: ΔRev.(t) = B0 + B1*Δ US_E_3M30Y_SWP_LOG(t) + u(t). Standard errors in parenthesis. * p<.1, ** p<.05, ***p<.01. Newey-West standard errors with a maximum of 4 lags. T = 59 observations, 2010 Q2 – 2024 Q4 inclusive. Revenue is in $MN.

[21] See the Appendix for a derivation. Modelers estimate the first-differenced regression to use stationary covariates and a stationary dependent variable. If the differenced model is valid, the levels specification is as well.

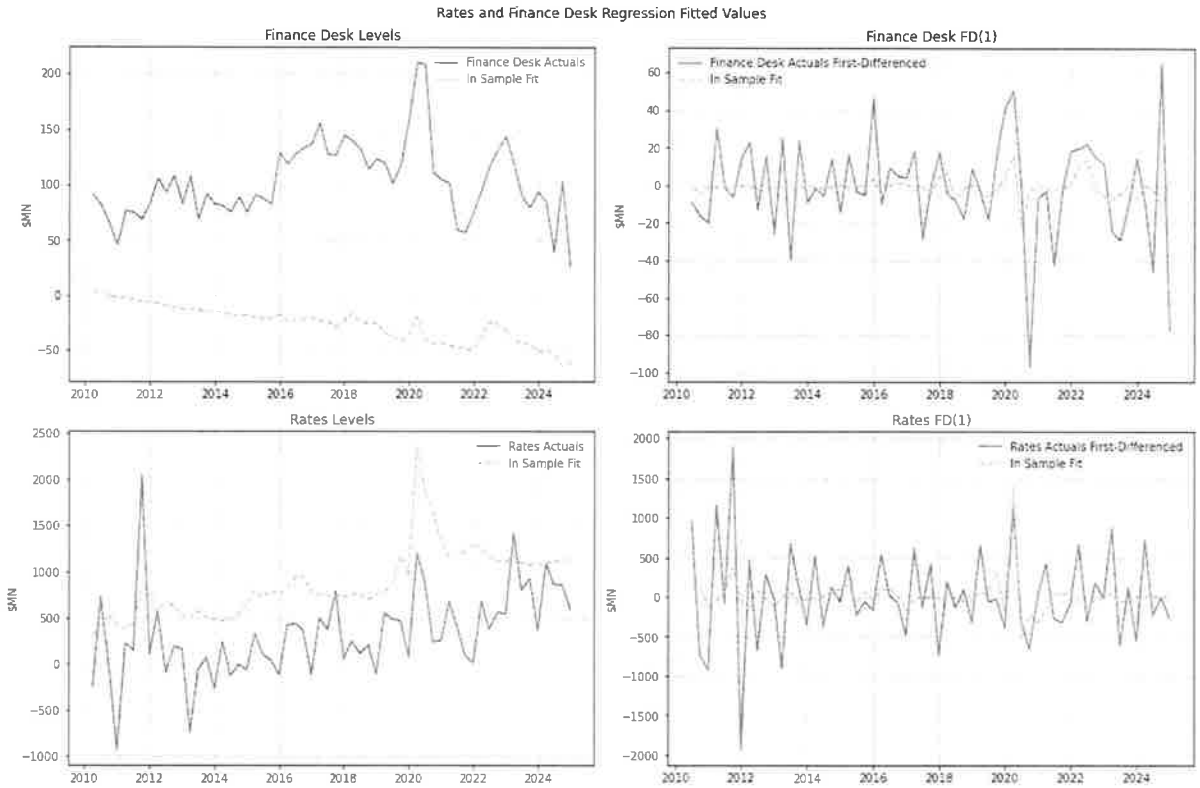
Both models perform poorly. Neither intercept is statistically different from 0, and the Finance Desk's intercept is negative, suggesting a downward trend. Only Rates has a significant covariate at a 5% level. Applying the above estimated parameters to data in levels produces implausible jump-off values:

MEV Approach Jump-Off Values, Finance Desk and Global Rates

Q_0	Finance Desk	Global Rates
Actual	$26.68 MN	$591.83 MN
MEV Estimated	-$61.24 MN	$1,135.13 MN

Q0 is 2024 Q4. MEV estimated jump-off values are from applying regression coefficients to MEVs and levels and a time trend (T=59).

despite visually plausible in sample fits for the change regressions, shown below:



OLS prediction intervals for forecasts in levels are too wide to be practically useful and introducing variation from the choice of jump off only widens them. The Finance Desk and Global Rates examples are not a *conceptual* indict of the levels macroeconomic variable regression approach. These examples merely show that certain desks may have better forecasting regression models, and thus be more suitable to the levels macroeconomic variable regression approach for selecting a jump-off.

Hodrick-Prescott Filter

The HP filter is a canonical approach, and often the default method to separate trend from cycle in macroeconomic variables like unemployment and GDP. It utilizes a single parameter based upon time series frequency (annual, quarterly, or monthly) and produces visually appealing smoothed series. Let y(t) be a variable with T observations, g(t) a smooth series like y(t) that solves:

$$\min_{g_{t=1\ldots T}} \sum_{j=1}^{T}(y_t - g_t)^2 + \lambda \sum_{t=1}^{T}[(g_t - g_{t-1}) - (g_{t-1} - g_{t-2})]^2$$

where λ is a smoothing parameter. Hodrick and Prescott (1997) suggest standard values for λ based upon the observation frequency: quarterly, 1600, monthly, 129,600, and annual, 6.25. The objective function minimizes the difference between smoothed and actual while penalizing the second difference, and therefore curvature, of the smoothed g(t). At the limit for λ, g(t)'s second difference goes to 0 and g(t) is effectively a regression of y(t) on a time trend. The problem does have a closed form solution, but it requires inverting a large sparse matrix. Instead, decompose y(t) into a trend, g(t), and cycle, c(t):

$$y_t = g_t + c_t$$
$$g_t = 2g_{t-1} - g_{t-2} + v_t$$

where the cycle is a stationary and unobservable residual component. Let v(t) be the second differenced trend. Assume further that v(t) and c(t) are uncorrelated white noise processes. Let λ be the ratio of the variances[22] between c(t) and v(t): $\lambda = \sigma^2_c/\sigma^2_v$. The trend-cycle component can be represented in state space form, and g(t) computed using the Kalman smoother:

$$x_t = \begin{bmatrix} g_t \\ g_{t-1} \end{bmatrix}, A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}, C = (1\ 0)$$
$$w_t = \begin{bmatrix} v_t \\ 0 \end{bmatrix}$$
$$x_{t+1} = Ax_t + w_t$$
$$y_t = Cx_t + e_t$$

for observed vector y(t), unobservable state vector x(t), state disturbance w(t), and measurement error e(t). The HP Filter has two primary drawbacks. First, the HP filter's objective is to produce a cyclical component that exhibits some discernible pattern. The cycle in the HP filter refers to business cycles, which are not white noise, but the jump-off exercise's objective is to isolate economically uninformative white noise[23]. Second, there exists recent academic evidence strongly discouraging the use of the HP Filter. In "Why You Should Never Use the Hodrick-Prescott Filter", Hamilton (2017) makes 3 arguments:

Objection 1: The HP filter is presumptively appropriate for economic time series, such as stock prices, long term interest rates, etc., typically considered random walks. If the HP filter doesn't pass the threshold of adequately separating trend from cycle in a simulated random walk, it shouldn't be used as an all-purpose approach. First-differencing a random walk produces noise, however Hamilton demonstrates that HP cycles estimated on random walks have meaningful predictability. However,

[22] Hodrick and Prescott note that results from the state space representation are predicated on distributional assumptions and are highly sensitive to the choice of smoothing parameter. They argue that formulating the smoothing parameter as a ratio of variances incorporates prior knowledge of the macroeconomic time series examined and therefore facilitates an empirically grounded choice of parameter.

[23] If the state space representation is the true data generating process for a time series y(t), in which measurement error e(t) = y(t) – g(t) = c(t), it should be the case that the estimated cycle c(t) = y(t) - $\tilde{g}(t)$ is white noise. Hamilton argues that this is a deficit to the HP filter, because users of the HP filter expect economically meaningful patterns in the cycle c(t). For the jump off exercise, however, white noise c(t) is desirable.

A) Revenue for markets desks clearly is not a random walk. First-differencing revenue shows serial correlation is still present, and the macroeconomic variable models which explain contemporaneous variation exhibit the requisite statistical significance and goodness of fit to reject the RW hypothesis.
B) In practice, residuals (actual minus smoothed) generated from using the HP filter exhibit among the lowest serial correlation among smoothing/denoising methods, and *lower* autocorrelation than that present when first differencing revenue actuals. In contrast, Hamilton cites the autocorrelation of the HP cyclical component of stock prices and real consumption spending in levels as much *higher* than autocorrelation in actuals.

Objection 2: The HP filter is two-sided. By construction, the trend at t is a function of future realizations at t+1...T. The minimization problem chooses a trend g(t) using all data, but in real time future values are unknowable: "And the end-of-sample value obtained from the procedure is actually quite different from what the researcher is seeing in the middle of the sample."(Hamilton 9). The point is moot for the purpose of obtaining Q_0 because Q_0 occurs at the end of the sample[24]. Motivation for the jump-off just is answering the question of what's the trend at the end of the sample with the benefit of hindsight.

Objection 3: Prescribed λ values do not cohere with actual economic data. λ can be estimated from the data by assuming normally distributed innovations (v(t), c(t))', using the Kalman filter to compute the likelihood on the observed sample y(t), and solving for the likelihood maximizing variance values. Hamilton finds λ values for quarterly macroeconomic series an order of magnitude smaller than HP's suggested 1600, such as 9.6 for GDP and 0.25 for the Federal Funds Rate. However,

A) Optimal λ, under modest parametric assumptions, can be determined empirically using the time series of revenue actuals per desk. The tradeoff of using MLE to obtain λ is that it negates the filter's advantage of not requiring hyperparameter fine tuning.
B) Hodrick-Prescott's choice of λ = 1600 for quarterly data supposes that a large intra-quarter change in the cyclical component is 5%, and the trend component, 0.125%, implying $\lambda = (5/0.125)^2 = 1600$. Such a supposition for markets desks, determined in consultation with the business, is no more subjective than the business prescribed jump-off values currently used.

Identifying the appropriate λ per desk is the key challenge for the HP filter. Unlike smoothing parameters in other methods, λ has a specific economic interpretation – the variance of the inverse signal to noise ratio – and the HP filter's output is predicated on a valid λ. Cross-validation won't suffice. I first assess whether the actual variance ratio of the trend and cycle produced by the default λ = 1600 for quarterly data aligns. I apply the HP filter with λ = 1600, compute the sample variance of the second differenced trend and sample variance of the cycle for each desk. If the actual λ differs from 1600, then the assumptions of v(t) and c(t) as serially uncorrelated white noise or the presumed λ is wrong. Second, for each desk I estimate the trend g(t) and compute the cycle c(t) from y(t) - $\tilde{g}(t)$ using a cubic spline[25]. I compute the requisite variances to establish an empirical prior for λ. Finally, I use maximum likelihood estimation to infer λ from the data. I present the different estimates in the table below:

[24] The concern is valid and precludes using the HP smoothed series as a dependent variable in a contemporaneous macroeconomic variable regression. The dependent data would reflect future observations, thus the regression in practice is of revenue at T on lagged covariates at t < T. Instead, use a one-sided HP filter, in which the objective function's smoothed value of g(t) only accesses y(t) actuals from 0 ... t.
[25] Cubic smoothing splines are piecewise continuous functions constituted by cubic polynomials. I use them because the trend must exhibit non-linearity for the variance of its second difference to be >> 0. A linear trend produces a v(t) with near zero variance and wildly high λ.

HP Filter λ Estimates

Desk	HP λ = 1600			Cubic Spline			MLE
	$\hat{\sigma}_c^2$	$\hat{\sigma}_v^2$	$\hat{\lambda}$	$\hat{\sigma}_c^2$	$\hat{\sigma}_v^2$	$\hat{\lambda}$	$\hat{\lambda}$
Commodities	6,920.515	0.146	47,420.934	5,671.699	5.769	983.186	22,019.707
DMCT	8,595.199	0.153	56,322.850	6,723.453	13.372	502.807	37,914.304
DM SP Trad.	41,654.786	0.332	125,373.408	36,285.007	32.279	1,124.095	95,491.467
EMCT	1,571.908	0.009	166,070.210	1,400.470	0.987	1,418.878	47,469.211
Equity Admin	5,474.736	0.236	23,229.272	4,484.745	3.570	1,256.317	41,557.550
Equity Cash	2,589.255	0.030	86,060.996	1,594.540	13.912	114.612	24,228.289
Equity Derivatives	28,838.377	0.504	57,233.142	25,920.284	14.348	1,806.499	41,030.830
F&S	6,126.718	0.085	72,047.611	1,805.062	786.021	2.296	50,983.290
FDC - Clearing	157.324	0.008	18,647.800	123.140	0.125	985.350	131,978.977
FX	23,955.527	0.485	49,348.584	15,205.139	211.640	71.844	77,298.432
Finance Desk	698.854	0.024	29,477.988	170.718	61.053	2.796	4,955.046
G10 MT	14,683.980	0.336	43,676.506	10,481.411	47.470	220.801	47,425.315
GSP Other	3,203.616	0.187	17,090.824	532.890	841.716	0.633	349.763
Global Rates	172,298.986	1.039	165,832.454	152,233.844	128.014	1,189.193	36,724.917
LMT	17,761.399	0.951	18,686.021	4,603.607	353.234	13.033	8.830
Other	13,504.283	0.913	14,795.877	1,690.952	9,245.940	0.183	76,523.755
Prime Finance	2,102.377	0.083	25,224.848	1,744.018	1.664	1,048.372	11,138.737

Hodrick-Prescott Filter λ values per markets desk, computed using revenue actuals from 2010 Q1 – 2024 Q4, inclusive (T=60). HP Filter separates revenue y(t) into a trend, g(t), and cycle, c(t): $y(t) = g(t) + c(t)$, where $v(t) = \Delta^2 g(t)$ and $\lambda = \sigma_c^2 / \sigma_v^2$. Columns 1-3 report sample variances of the cycle and second differenced trend where the trend and cycle are generated from an HP filter with the default λ = 1,600 for quarterly data. Columns 5-7 report the same where the trend and cycle are generated from a cubic spline, with a non-negative spline penalty parameter chosen using LOOCV. Column 7 reports λ estimated from the data using maximum likelihood, using spline λ as the optimizer's starting point.

Using Hodrick-Prescott's λ = 1600 to smooth market desks revenues is wrong. If it were correct ex-ante, the figures in column 3 should be close to 1600, but they are 10-100x larger owing to the cycles' high sample variance. The baseline cubic spline's trend and cycle produces empirical λ's that are much closer to 1600 for certain desks, such as EMCT and Equity Derivatives, and some closer to Hamilton's MLE estimates for various quarterly macroeconomic series, which range from 0.023 to 9.596. Maximum likelihood estimates are often 10x the prescribed 1600. Only Finance Desk, at λ = 4,955, is the right order of magnitude. The high MLE smoothing parameters suggest that the HP filter, when properly applied, will produce highly smoothed series, as opposed to the visually pleasing results with λ = 1600.

Absent valid, heterogenous λ values, the HP filter does not isolate a residual cycle. Usage of the HP filter consequently is contingent on valid λ values per desk.

Hamilton Filter

Hamilton (2017) proposes an alternative to the Hodrick-Prescott filter, to answer the question of how the value of a possibly non-stationary series at some future date, t+h, differs from the expected value at t+h, given only data observed through time t. Let y(t) be the series, and project y(t+h) on a constant and the p most recent values of y as of date t. In the population case, using p=4, the formulation is:

$$y_{t+h} = \beta_0 + \beta_1 y_t + \beta_2 y_{t-1} + \beta_3 y_{t-2} + \beta_4 y_{t-3} + v_{t+h}$$

Hamilton prescribes h = 8 for quarterly data, h = 24 for monthly data, and h = 2 for annual data. Forecasting 2 years ahead coheres with the horizon of a standard business cycle, over which the primary source of error is cyclical factors like recessions and recoveries from economic downturns. For data with seasonality, Hamilton argues p and h should be integer multiples of the number of observations in a year. Regression residuals provide a transient component, the cycle, and are stationary provided that y(t) is at most I(4). I verify that 4th differenced revenue is stationary for all desks. The residuals are:

$$\hat{v}_{t+h} = y_{t+h} - \hat{\beta}_0 - \hat{\beta}_1 y_t - \hat{\beta}_2 y_{t-1} - \hat{\beta}_3 y_{t-2} - \hat{\beta}_4 y_{t-3}$$

Lagged values comprise the signal. Advantages are as follows. First, the filter has minimal assumptions of linearity and the stationarity of the fourth differenced y(t). Second, the filter is one-sided. It uses lagged observations, so the smoothed series can be used as a dependent variable in contemporaneous regressions. Third, it is largely model free. Hamilton argues it achieves the objectives of the HP Filter with none of the drawbacks. However, there are several disadvantages. First, the attraction of a one-sided filter for regression modelling isn't pertinent for estimating jump-off values. In the latter case, it is desirable to use all available data, as a two-sided filter does. Second, the filter is in essence the 4QMA approach currently in use, but with OLS determined instead of equal weights. Set h = 0, $\beta_0 = 0$, $\beta_{1...}\beta_4$ to 0.25, and the Hamilton Filter takes 4QMA's exact form. Deviating from p = 4 and h = 8 demands another calibration exercise or CV procedure. Third, the filter's goal is to decompose a time series into a trend and economically meaningful cycle. The cyclical component may not be pure white noise, but for jump-off analysis, a pure noise cycle is desirable.

Scoring Function

To assess the performance of smoothing methods for each desk's revenues, I develop a custom scoring function. The optimal smoothing method removes the source of 'noise' outlined earlier, such as idiosyncratic revenue shocks and measurement error, but the determination of what constitutes noise can only be done ex-post, with the benefit of historical data. If model developers had clean historical revenues, the jump-off analysis would be moot[26]. To avoid favoring one model over another, I eschew common objective functions. For example, the criteria of minimal MSE relative to actuals would select the method that produces the least smoothing. Incorporating the difference between actual and smoothed in levels into the criterion favors the HP filter, and so forth. I use a weighted average of broad statistical criteria to favor the desirable properties of a smoothed/denoised series relative to actuals without relying on specific domain knowledge. Let g(t) be the candidate smoothed series, and λ_i are non-negative weights that sum to 1. The optimal smoothing method minimizes the following criterion:

$$g^*(t) : \min_g \lambda_1 Q_e + \lambda_2 s_e + \lambda_3 Q_{y,g} + \lambda_4 \mathrm{FD}_{\text{t: Inflection}} + \lambda_5 \mathrm{SD}_{\text{t: Local Extrema}}$$

Define residuals e(t) as actual minus smoothed. To capture properties of white noise, the criterion penalizes residual serial correlation and time-varying volatility. White noise is mean zero, serially uncorrelated, and has constant variance[27]. I include the cross-correlation function, computed between actual and smoothed, to penalize smoothed series that produce spikes at lags or leads of the actuals. This component penalizes the Slutzky-Yule effect, for example, associated with MA smoothing. A smoothed series should show large revenue spikes at the same quarter as the original series. To favor smoothness, I include the sum of squared second differenced-smoothed values computed only at the time indices of the actual's local extrema. The second difference is the discrete time analogue to the second derivative of a continuous function and therefore represents curvature. I also include the L1 norm of first-differenced smoothed minus first-differenced actuals to favor smoothed series with the same directional movements at time indices of actual's inflection points. The components are below:

$$\text{Residual Serial Correlation} : Q_e \equiv \sum_{k=1}^{h} \hat{\rho}_k^2$$

$$\text{Residual Time-Varying Volatility} : s_e \equiv ||\hat{\sigma}_{e,i}^2 - \hat{\sigma}_e^2||_2^2$$

$$\text{Smoothed vs. Actual CCF} : Q_{y,g} \equiv \sum_{s \neq t} \hat{\rho}_{y,g}^2(s,t)$$

$$\text{Smoothed vs. Actual FD (Slope)} : \mathrm{FD}_{\text{t: Inflection}} \equiv ||\Delta y_t - \Delta g_t||_1$$

$$\text{Smoothed SD (Curvature)} : \mathrm{SD}_{\text{t: Local Extrema}} \equiv ||\Delta^2 g_t||_2$$

I normalize each score component by dividing it by the sum of components across methods, obtaining a percentage. Components are defined such that lower scores for each are better. Adjusting the weights for each component[28] permits modelers to prioritize desired features of the smoothed series. Setting $\lambda_5 = 1$, for example, would select the method with a fit closest to a straight line. Setting $\lambda_2 = 1$ would select

[26] Simulation involves generating a signal, adding noise, and comparing each smoothing method's estimated signal to the known actual signal. Using simulation to score smoothing methods would select the method best aligned to the structure of the signal because the smoothing/denoising methods include common data generating processes for a time series, such as ARIMA and trend/seasonality.

[27] I compute the sum of squared deviations between rolling residual variance, using window size w=T^0.5, and full sample residual variance. For T=60, w=8, capturing 2 years of quarterly data. If overall residual variance is low, and rolling variance is low, so is s(e), the squared L2 norm of deviations between sample variance of subgroup i's residuals and sample variance of residuals e(1)...e(T), where i a size w subset of T.

[28] Weights for defined components get rescaled (e.g. $\lambda_i = 0.20 \rightarrow \lambda_i = 0.25$) for undefined score components (e.g. LB stat for residuals of 0).

the smoothing method that does no smoothing, resulting in residuals of 0 with volatility 0. By design, the scoring criterion is flexible, and a method's score reflects the choice of weights[29].

The table below shows the top five smoothing methods over all desks per weighting scheme[30]. Each scheme emphasizes a particular aspect of the scoring function. The first row treats each component equally, the second prioritizes the absence of residual serial correlation and doesn't consider co-movement between smoothed and actual, the curvature of smoothed, or the cross-correlation between smoothed and actual. The last row is a balanced weighting scheme, favoring residual properties.

Smoothing/Denoising Method Rankings

Score Component Weight					Method Ranking				
Serial Corr.	Resid. Vol.	Slope	Curvature	CCF	1st	2nd	3rd	4th	5th
0.2	0.2	0.2	0.2	0.2	HP Filter	FFTCV	Lowess	Wavelet	Cubic Spline
0.8	0.2	0.0	0.0	0.0	Auto SARIMA	HP Filter	Lowess	FFTCV	Reg.
0.5	0.5	0.0	0.0	0.0	HP Filter	Lowess	FFTCV	Reg.	Wavelet
0.6	0.2	0.0	0.0	0.2	Auto SARIMA	Lowess	HP Filter	FFTCV	Reg.
0.0	0.0	0.4	0.4	0.2	HP Filter	FFTCV	FFT1	Lowess	Cubic Spline
0.0	0.0	0.33	0.33	0.33	HP Filter	FFT1	FFTCV	Lowess	Wavelet
0.4	0.3	0.10	0.10	0.10	FFTCV	Lowess	HP Filter	Wavelet	Reg.

Ranking of smoothing methods per weighting scheme. For each smoothing method and desk, and using lambda weights in columns 1-5, compute $\lambda_1 Q(e) + \lambda_2 s(e) + \lambda_3 Q(y,g) + \lambda_4 FD + \lambda_5 SD$, a weighted average of residual serial correlation, time-varying residual volatility, cross-correlation between smoothed and actual, absolute difference between the change in actual and change in smoothed (slope), and L2 norm of second differenced smoothed (curvature). For each desk, rank order the methods, e.g. for commodities: HP Filter, Lowess, Reg., etc. Finally, rank order methods across desks by summing ranks and sorting from least to greatest.

Equal weighting favors the HP Filter and FFT with cross-validation. Minimizing residual serial correlation favors the Auto Arima and trend-seasonality regression approaches. Favoring the visual elements of smoothing, namely slope, curvature, and co-occurrence of peaks, isolates the HP filter and signal processing methods. For all but one weight vector considered, the HP filter enters the top 5, and for all but three vectors considered, FFT with cross-validation is in the top 5. When the scoring function emphasizes the time series properties of white noise, it favors the Auto SARIMA and regression methods because by construction they isolate white noise from signal. When the scoring function prioritizes the criteria associated with a visually plausible smoothed series, the HP filter, signal processing, and cross-sectional smoothing methods rank highly.

Given the signal plus noise framework, the appropriate weighting scheme should weight residual serial correlation most. Under such schemes, Auto SARIMA ranks 1st.

[29] A natural test is to use the fitted jump off values as static forecasts over a held back or rolling test period, and compute forecasting errors per method, horizon, and desk. I report such results in the Appendix. The small sample size and overlap precludes drawing meaningful insights.
[30] See the Appendix for complete results. Weights are non-negative and sum to 1.

Recommendation

Based on the signal plus noise decomposition of revenue actuals, scoring function and predictive test results, and the constraints of simplicity, interpretability, and usability, **I recommend complementing the 4Q average and decision tree framework with an Auto SARIMAX fitted to the time series of revenue actuals for each PPNR markets desk to obtain jump off values at Q_0, in conjunction with the trend/seasonality regression model, as business tools**. These methods do not require any modeler intervention or nuanced cross-validation procedures for hyperparameter tuning, preserve the time series properties of the actuals data, and should be feasible to get through model validation. I discuss the merits and demerits of each smoothing/denoising method below.

Signal Processing Methods:

- Fast Fourier Transform with Cross-Validation: Using a balanced weighting scheme for the custom scoring function, and per the holdback and rolling tests of predictive power, FFT with CV is the optimal denoising method. The combination of preliminary detrending and cross-validation for the optimal cutoff frequency results in superior performance compared to FFT using the 1st periodogram peak as a cutoff and Wavelets. However, the complexity of the transformation and non-intuitive nature of signal reconstruction from the frequency domain – there is no sense of *when* key frequencies occur, only that they exist – make validation and usage practically challenging.

- Wavelet Analysis offers greater potential than FFT by representing a signal in the time and scale domain and capturing the local properties of the signal. When applied to quarterly market desk revenue actuals, it performs worse than FFT with CV because the threshold based denoising simply does not have enough levels to precisely remove noise from the signal. FFT with CV offers granular cutoff frequencies per desk, so its aggregate scores are higher than that of Wavelets. The choice of Wavelet function and level for thresholding demands more complicated analysis and procedures than the method agnostic cross-validation procedures I use for FFT and Lowess. Wavelets suffer from the same interpretability, complexity, and usage constraints as Fourier transforms. Wavelets may perform better using a larger sample of more frequent data and are worth further study.

Filtering Methods:

- HP Filter: It is inadvisable to use the Hodrick-Prescott Filter, despite the visually compelling smoothed series it produces and the top performance in the custom scoring framework across score component weighting schemes. First, the academic evidence against its usage is strong. Hamilton's 2017 critique features more prominently than the original paper, and the exculpatory nuances in Hamilton's objections are difficult to explain away. Second, results are contingent on a heterogenous smoothing parameter, estimation of which requires a complicated procedure. Although the HP filter is interpretable, derivations for smoothing parameter estimation are exhaustive. Using valid parameters instead of the default 1,600 produces flat smoothed series.

- Hamilton Filter: There is no compelling reason to use the Hamilton filter. It is, in effect, a regression based 4Q average and performs poorly for the same reasons as the 4Q approach.

Regression Approaches:

- Trend/Seasonality Regression: The trend/seasonality regression aligns with the signal plus noise framework. The noise is the OLS error term, and the signal is the sum of linear and quadratic trends, and quarterly indicators. By conventional measures of goodness of fit and statistical significance, the regression models perform well enough for developers to have confidence in the fitted Q_0, despite middling performance and predictive power. Minimally, the estimates of quarter specific intercepts and time trends inform jump off analysis. The model is simple, interpretable, can be fit in Excel, and by quantifying seasonal and trend effects can be used as a business tool.

- Levels Macroeconomic Variable Regression: If the forecasting model is correct, the MEV regression in levels is the only valid model for the jump off. The quality of estimated Q_0 depends entirely on the quality of the forecasting model. The examples using Finance Desk and G10 rates show wildly implausible estimates. For large enough estimated MEV coefficients, the jump off can even be negative. I cannot recommend this approach.

- Lowess: An advantage of local regression is that it's well suited to non-stationary time series by virtue of using local moments of the series for estimation. It two principal deficits. First, fitted Q_0 for many desks is close to actual Q_0, regardless of fraction, because the estimate at the end of the sample can only half of the observations (those immediately prior) in the bandwidth. Second, the CV procedure dictates fraction close to 1 for several desks, resulting in effectively linear trend fits, and fractions close to 0 for several desks, resulting in no smoothing at all.

Time Series Methods:

- Auto SARIMAX: The most conventional time series method is closely aligned with the signal plus noise framing, straightforward to defend, implement, and interpret the results of, and tends to produce residuals empirically most akin to the theoretical noise (no serial correlation, constant variance). The method performs well when the scoring criteria favor the properties of white noise, and less so when it favors visual components like curvature. SARIMA models explicitly account for the temporal structure of the data, seasonality, and offer an extension for exogeneous macroeconomic variables. Modelers can take the covariates from the change regression model, fit a regression, estimate an ARMA for the residuals, and re-estimate the entire SARIMAX using MLE. Using a SARIMAX ensures logical consistency with the forecasting model and generates quality estimates of Q_0 ,net of exposure to macroeconomic covariates.

Baseline Methods:

Cubic Spline and Moving Average methods should not be used. Neither coheres with the framing of revenue as signal plus noise.

- Cubic Spline: A cubic spline is devoid of any economic content and disregards the time series structure of the data. Smoothed series are visually compelling, but the spline ranks last or close to it when the score function weights residual serial correlation and volatility.
- Moving Average: Using a moving average *induces* serially correlated residuals, where serial correlation increases at higher lags and can exceed serial correlation in the revenue actuals. MA performs well in holdback and rolling predictive tests merely by capturing local trends.

The table below shows jump-off values for Q1 2025 forecasts per desk and method.

Q4 2024 Jump-Off Values ($MM)

Desk	Q4 2024 Actual	Auto Arima	Cubic Spline	FFT1	FFTCV	HP Filter	Hamilton	Lowess	MA	Reg.	Wavelet
Commodities	$306.03	$247.36	$241.32	$202.34	$287.01	$283.22	$296.56	$233.09	$241.27	$262.73	$165.17
DMCT	$76.76	$37.66	$83.85	$121.93	$125.04	$85.59	$108.16	$88.93	$104.48	$40.72	$107.31
DM SP	$194.32	$70.16	$231.32	$427.04	$203.36	$197.47	$269.55	$168.5	$246.6	$46.09	$336.65
EMCT	$64.52	$7.5	$64.16	$94.59	$43.25	$48.37	$42.64	$45.65	$77.38	$3.23	$74.87
Equity Admin	-$148.0	-$162.95	-$169.39	$78.72	-$125.15	-$168.55	-$77.89	-$173.02	-$160.77	-$181.09	-$71.55
Equity Cash	$524.17	$254.93	$392.5	$297.01	$300.31	$282.03	$261.43	$249.49	$310.87	$263.82	$286.08
Equity Derivatives	$171.0	$294.2	$447.74	$384.54	$567.2	$507.02	$251.35	$529.21	$556.25	$421.69	$387.05
F&S	$798.27	$726.19	$794.0	$593.82	$755.42	$746.66	$653.93	$791.98	$766.84	$739.5	$605.73
FDC - Clearing	$140.6	$142.58	$151.08	$107.74	$150.46	$144.41	$118.47	$140.6	$145.95	$139.18	$107.12
FX	$1,102.78	$996.55	$1,067.6	$1,025.89	$1,049.58	$1,039.9	$980.36	$1,109.95	$1,022.63	$995.74	$1,015.39
Finance Desk	$26.68	$104.23	$44.42	$86.06	$84.67	$68.6	$108.72	$26.68	$63.73	$71.21	$75.39
G10 MT	$346.34	$455.39	$445.35	$302.96	$465.66	$410.7	$307.3	$449.84	$443.58	$394.43	$380.5
GSP Other	-$87.39	-$87.01	-$90.01	$24.94	-$55.84	-$92.67	-$73.1	-$97.94	-$95.05	-$62.13	$22.12
Global Rates	$591.83	$521.35	$803.68	$461.84	$815.18	$842.35	$477.5	$762.9	$848.59	$584.2	$604.9
LMT	$1,161.69	$1,174.83	$1,160.25	$835.05	$1,231.3	$1,235.69	$868.8	$1,142.34	$1,230.21	$1,113.65	$958.05
Other	-$148.25	-$125.01	-$147.6	$26.62	-$59.99	-$147.22	-$41.01	-$148.25	-$160.0	-$107.76	-$45.79
Prime Finance	$299.84	$313.14	$318.31	$270.78	$309.38	$301.79	$290.42	$325.15	$313.04	$266.38	$253.01

Jump off values per desk and smoothing/denoising method. Column 1 shows Q4 2024 actual values. All figures in $MM.

Appendix

Code and Output:

JO_Analysis_vF.html

JO_Analysis_vF.ipynb

Data:

pearl_output.parquet

Examples of Analytic Solutions:

JO_Framework.xlsx

Smoothed/Denoised Actuals Per Method:

smoothed_jo2.xlsx

Plots Per Method:

Smoothed_Fits.zip

Analytic Sensitivity Measures

Below are derivations for the sensitivity of revenue forecasts, in levels, for each type of dependent variable transformation. Let delta be a perturbation to the jump off value Y_0.

NOD1

$$
\begin{aligned}
Y_t - Y_{t-1} &= f(B, X_t, \ldots, X_{t-k}) + u_t \\
Y_t &= Y_0 t + \sum [f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\tilde{Y}_t &\equiv (Y_0 + \delta)t + \sum [f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\tilde{Y}_t - Y_t &= \delta t
\end{aligned}
$$

PCT1

$$
\begin{aligned}
\frac{Y_t - Y_{t-1}}{Y_{t-1}} &= f(B, X_t, \ldots, X_{t-k}) + u_t \\
Y_t &= Y_{t-1}\,[1 + f(B, X_t, \ldots, X_{t-k}) + u_t] \\
Y_t &= Y_0 \prod [1 + f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\tilde{Y}_t &\equiv (Y_0 + \delta) \prod [1 + f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\frac{\tilde{Y}_t}{Y_t} &= 1 + \frac{\delta}{Y_0}
\end{aligned}
$$

DP1

$$
\begin{aligned}
\frac{Y_t}{Y_{t-1}} &= f(B, X_t, \ldots, X_{t-k}) + u_t \\
Y_t &= Y_{t-1}\,[f(B, X_t, \ldots, X_{t-k}) + u_t] \\
Y_t &= Y_0 \prod [f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\tilde{Y}_t &\equiv (Y_0 + \delta) \prod [f(B, X_t, \ldots, X_{t-k}) + u_t] \\
\frac{\tilde{Y}_t}{Y_t} &= 1 + \frac{\delta}{Y_0}
\end{aligned}
$$

LND1

$$
\begin{aligned}
\ln\left(\frac{Y_t}{Y_{t-1}}\right) &= f(B, X_t, \ldots, X_{t-k}) + u_t \\
Y_t &= Y_{t-1} e^{f(B, X_t, \ldots, X_{t-k}) + u_t} \\
Y_t &= Y_0 e^{\sum [f(B, X_t, \ldots, X_{t-k}) + u_t]} \\
\tilde{Y}_t &\equiv (Y_0 + \delta) e^{\sum [f(B, X_t, \ldots, X_{t-k}) + u_t]} \\
\frac{\tilde{Y}_t}{Y_t} &= 1 + \frac{\delta}{Y_0}
\end{aligned}
$$

First-Differenced Revenue Properties

	Seasonality P-Value	Seasonality Present	LBP P-Value	Serial Correlation Present	ADF P-Value	Stationary	Cusum P-Value	Structural Break Present
Commodities	0.015	True	0.018	True	0.000	True	1.000	False
DMCT	0.000	True	0.000	True	0.000	True	0.991	False
DM SP	0.000	True	0.000	True	0.000	True	1.000	False
EMCT	0.000	True	0.000	True	0.000	True	1.000	False
Equity Admin	0.485	False	0.005	True	0.000	True	0.941	False
Equity Cash	0.000	True	0.002	True	0.132	False	0.927	False
Equity Derivatives	0.000	True	0.000	True	0.000	True	0.999	False
F&S	0.891	False	0.090	False	0.000	True	0.997	False
FDC -Clearing	0.001	True	0.007	True	0.000	True	1.000	False
FX	0.000	True	0.009	True	0.000	True	1.000	False
Finance Desk	0.038	True	0.689	False	0.000	True	0.958	False
G10 MT	0.738	False	0.000	True	0.000	True	1.000	False
GSP Other	0.345	False	0.078	False	0.002	True	0.521	False
Global Rates	0.000	True	0.000	True	0.000	True	0.984	False
LMT	0.007	True	0.279	False	0.000	True	0.996	False
Other	0.027	True	0.053	False	0.000	True	0.819	False
Prime Finance	0.003	True	0.018	True	0.000	True	0.939	False

Time series property of revenue actuals for PPNR markets desks. T=59, Q2 2010 – Q4 2024, inclusive. Ljung-Box test for serial correlation, Kruskal-Wallis test for seasonality, Augmented-Dickey Fuller test for stationarity, and CUSUM test on residuals of trend regression for structural break. All tests conducted at a 5% level.

Time Series Test Details

I perform Ljung-Box, Kruskal-Wallis, and Augmented Dickey-Fuller tests at a 5% level. I also use a CUSUM test for parameter stability. Test details are below.

Ljung-Box

A Ljung-Box test is a test of serial correlation in a sample time series. The null hypothesis is:

$$H_0 : \rho_1 = \ldots = \rho_m = 0$$

for a time series $X_{1:T}$ sample, and the alternative is:

$$H_a : \rho_i \neq 0$$

for any i in {1:m} where $1 \leq m \leq T$. The test statistic has a Chi-Square distribution:

$$Q_{\mathrm{LB}}(m) = T(T+2) \sum_{k=1}^{m} \frac{r_k^2}{T-k} \sim \chi^2(m)$$

for m lags and r_k the sample ACF of $\{X_t\}$. Under the null hypothesis, autocorrelations are zero. I set m using the frequency of the time series: m = 4 for quarterly data, and m = 12 for monthly data. This accords with the business intuition of the relevance of the trailing year's actuals.

Augmented Dickey-Fuller

The ADF test assesses whether a time series contains a unit root, meaning the auto-regressive polynomial has a root of 1 and therefore the series is non-stationary, using a regression:

$$\nabla x_t = \gamma x_{t-1} + \sum_{j=1}^{p-1} \phi_j \nabla x_{t-j} + W_t$$

where

$$\gamma = \sum_{j=1}^{p} \phi_j - 1$$

The null and alternative hypothesis are that the process is non-causal:

$$H_0 : \gamma = 0$$

and causal, respectively:

$$H_a : \gamma < 0$$

An ADF test requires two user decisions. The first is the restriction imposed on the underlying process X(t). Using an AR(1) for simplicity, the most general formulation is:

$$x_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \phi_1 x_{t-1} + w_t$$

Statsmodels implements the ADF test using flags for the restriction that correspond to:

$$
\begin{aligned}
\text{"c"} &: \beta_1, \beta_2 = 0, x_t = \beta_0 + \phi_1 x_{t-1} + w_t \\
\text{"ct"} &: \beta_2 = 0, x_t = \beta_0 + \beta_1 t + \phi_1 x_{t-1} + w_t \\
\text{"ctt"} &: x_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \phi_1 x_{t-1} + w_t \\
\text{"n"} &: \beta_0, \beta_1, \beta_2 = 0, x_t = \phi_i x_{t-1} + w_t
\end{aligned}
$$

where the choice of restriction bears meaningfully on the ADF tests size and power. For tests on revenue in levels, I use 'ct', for first-differenced revenue I use 'c', and for higher order differences I use 'n'. First-differencing a constant and linear time trend drops the constant and keeps the trend coefficient. Differencing again drops the trend coefficient. The second choice is how many lags to include in the regression. I use a maximum of 4 lags for quarterly data, 12 for monthly data, and have Statsmodels select the optimal lag using AIC.

Kruskal Wallis

A KW test is a nonparametric test of the differences between the distributions of groups, where the distribution is computed on ranks on observations in the group. It is analogous to an ANOVA test but doesn't presume normality. The test ranks the data and partitions the data into groups based upon period (e.g. separate groups for Q1, Q2, Q3, and Q4 observations), using the intuition that if seasonality is present, the distribution of values in one period should differ from those of other periods.

The null hypothesis is that the medians of each group are the same. The alternative is that the median for at least one group differs from the others. For T observations, k groups (k=4 for quarterly data), rank R(i) of the i'th observation and n(i) observations in group I, the test statistic:

$$
H = \frac{12}{T(T+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(T+1)
$$

is compared to a Chi-Square distribution with k-1 degrees of freedom. The test assumes groups are mutually independent, so I cite the test results from first-differenced data.

CUSUM

Refer to pp.7 of the Stata manual[31] for 'estat sbcusum' for a concise technical treatment of the test. For covariates x(t), I use a constant and time trend.

[31] https://www.stata.com/manuals15/tsestatsbcusum.pdf

Decision Tree Programmatic Output

Below is console output for the automated decision tree, using equity admin and FDC – Clearing. The flag 'Rep' is user set and corresponds to whether revenue at Q_0 is representative of the go-forward state of the business. The tuples show the average and average plus or minus 1 standard deviation for the prior 4 quarters, 3 years, and 5 years of revenue actuals, inclusive of Q_0.

Equity Admin

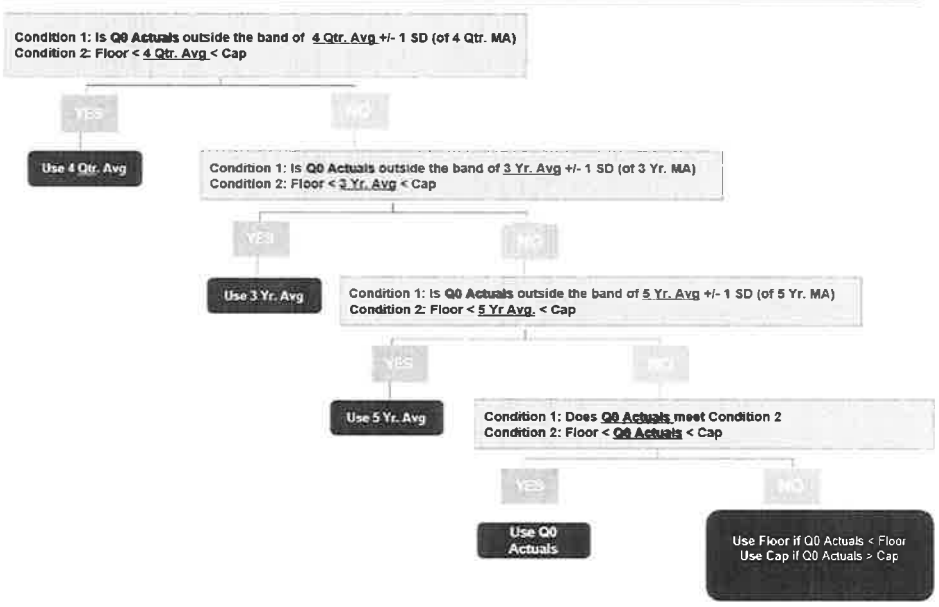
```
Decision Tree: Equity Admin
Freq: Q-DEC, Q0: -147.997669863, Rep:False
4Q: (-220.3326095158933, -160.774018934, -101.2154283521067)
3Y: (-173.17435688812034, -133.45998978733334, -93.74562268654634)
5Y: (-144.2273198952433, -112.56697406305, -80.90662823085668)
Does recent 4Q include strategic initiatives and represent go forward state?
-> No
Is 4Q Avg. within band of MT Trend (3Y Avg +- 1SD of 3Y MA)?
-> Yes
Is 4Q Avg. within band of LT Trend (5Y Avg. +- 1SD of 5Y MA)?
-> No: Use 5Y Avg.
```

FDC Clearing

```
Decision Tree: FDC - Clearing
Freq: Q-DEC, Q0: 140.60352522600002, Rep:True
4Q: (112.43520053867624, 145.94506298025, 179.45492542182373)
3Y: (102.30361386608769, 131.61784972983332, 160.93208559357896)
5Y: (95.94032897306548, 120.8245030475, 145.70867712193453)
Does recent 4Q include strategic initiatives and represent go forward state?
-> Yes
Is Q0 within band of ST Trend (4Q AVG +- 1SD of rolling 4QMA)?
-> Yes: FP&A/Bus. specific JO
```

The team has proposed a revised decision tree approach that incorporates floors and caps. However, the tree's terminal nodes are the same 4Q, 3-year, and 5-year averages:

Proposed Jump Off Framework (Decision Tree Approach)



FFT: Cross-Validation Procedure

Algorithmically, the cross-validation procedure to select the optimal cutoff frequency for FFT denoising:

Notation:

- X: discrete time signal
- N: length of X
- C: candidate cutoff frequency index, 1...N//4
- i: index of observation x[i], 0....N-1

Algorithm:

For each C:

For i = 0 ... N-1:

Let 'X modified' be X with x[i] removed

Insert interpolated value into X modified at [i]

if i = 0, the first index, use x[1], the second actual

if i = N-1, the last index, use x[N-2], the 2nd last actual

otherwise, use 0.5*x[i-1] + 0.5*x[i+1], the average of x[i]'s two neighbors

Smooth X modified using FFT with C

Compute the squared error between x[i] from X and x[i] from X modified

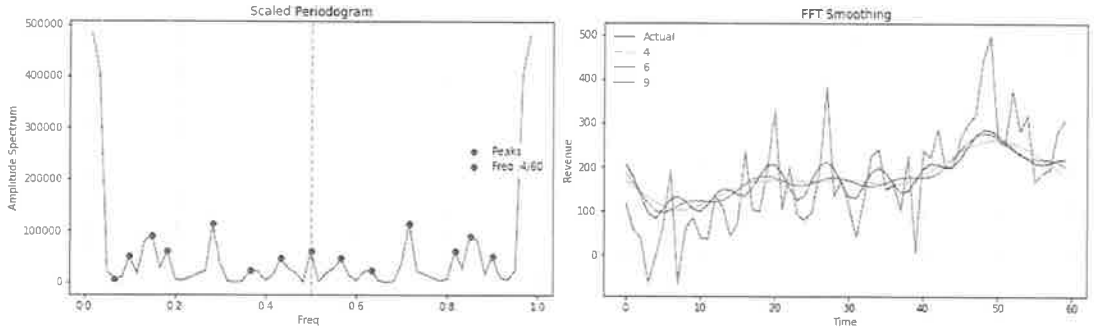
Average the squared errors over i, add a penalty term: 1000*C/N

Select the C which minimizes the MSE plus penalty term

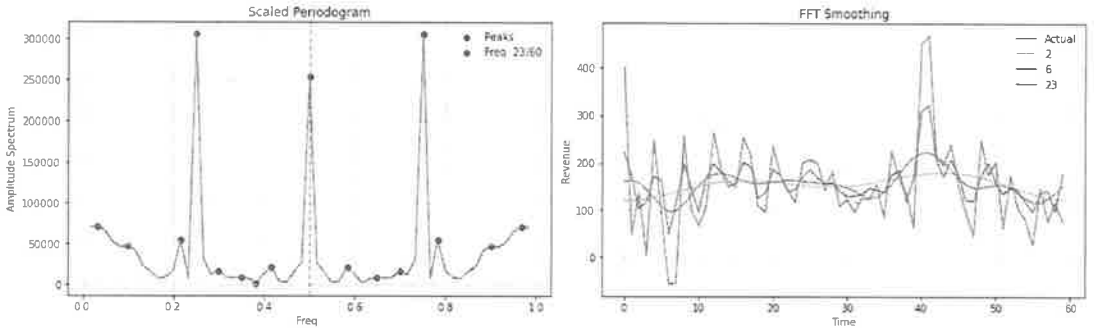
The penalty term increases as the cutoff frequency approaches half of the folding frequency. The range of cutoffs considered is 1, corresponding to frequency 1/N, up to N//4 - 1, which is half of the folding frequency using floor division. Bounding the cutoff range forces the algorithm to prescribe a lower frequency. I calibrated the penalty term weight 1000 based on trial and error, although it's tunable.

FFT: Desk Periodograms

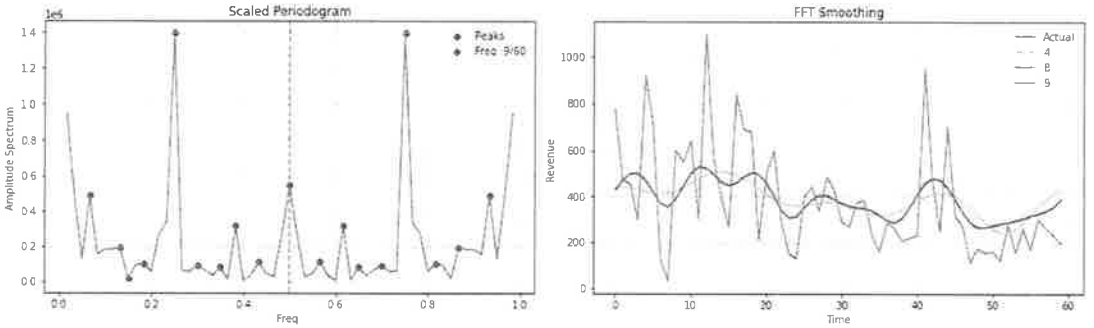
Commodities



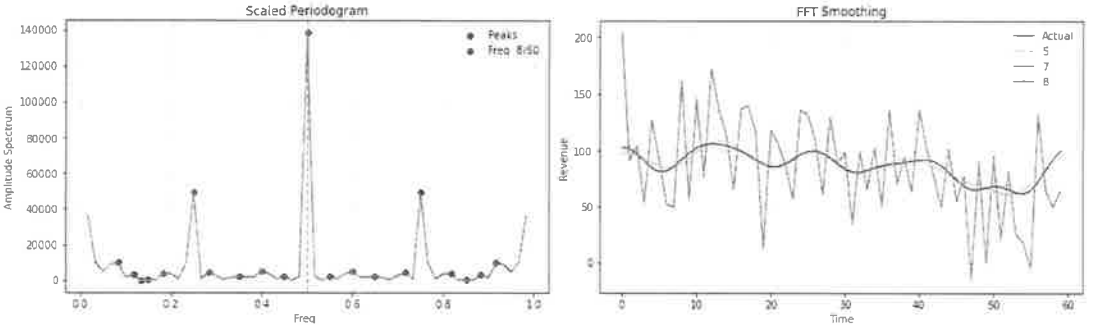
DM - Credit Trading



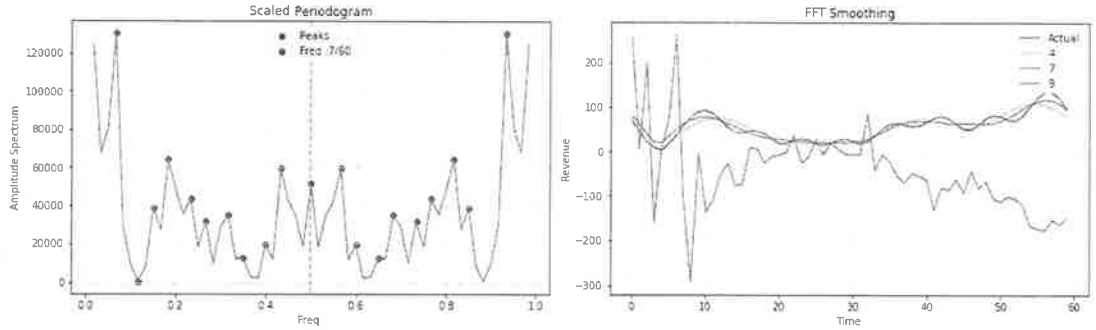
DM - Structured Products Trading



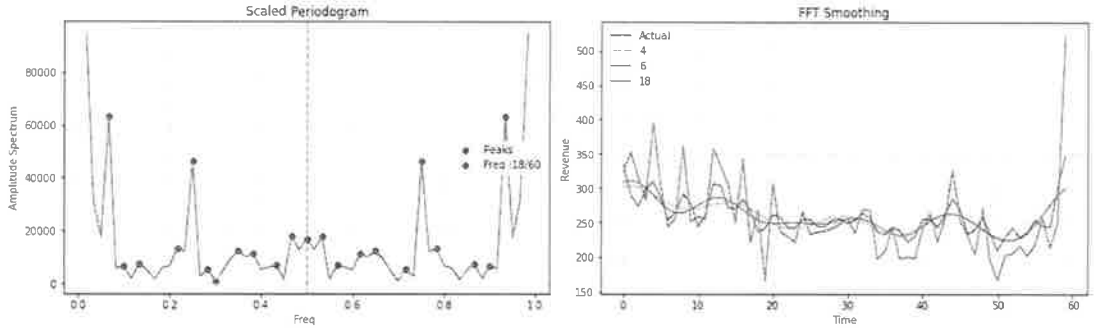
EMCT



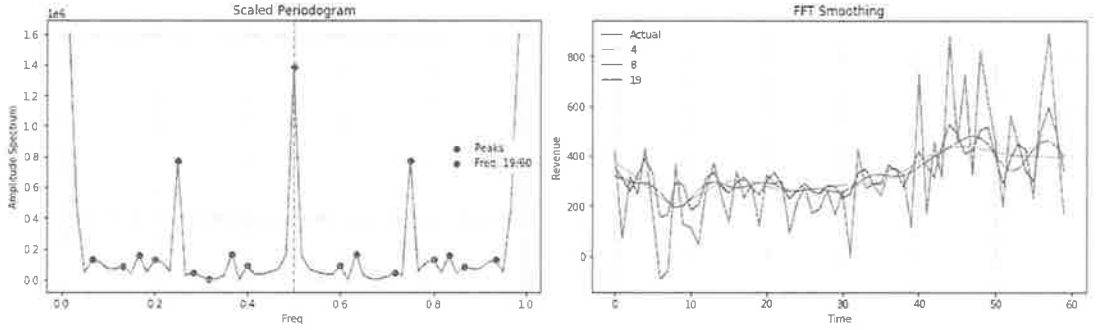
Equity Admin



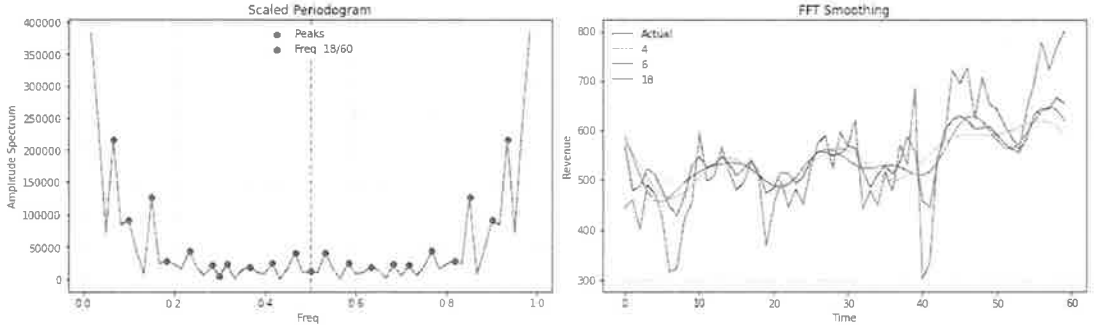
Equity Cash



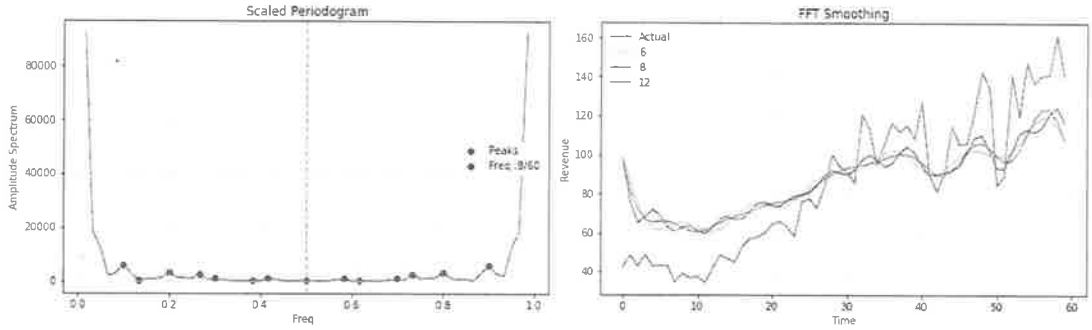
Equity Derivatives



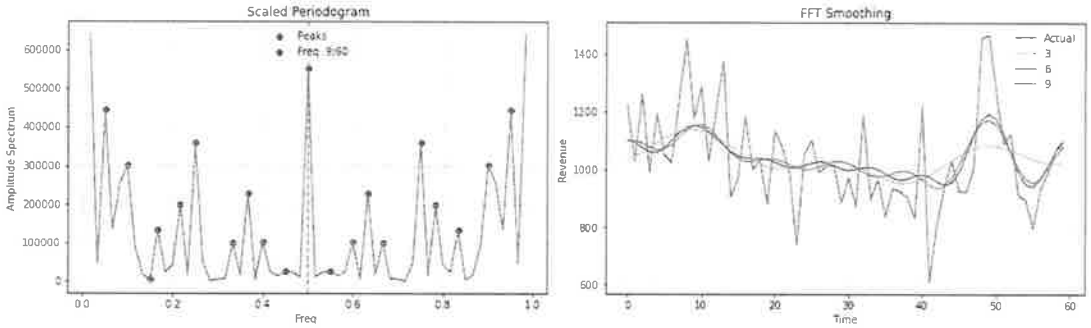
F&S



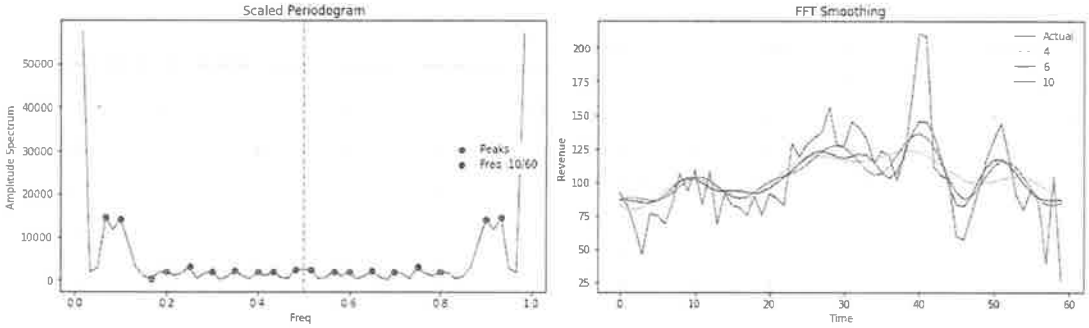
FDC - Clearing



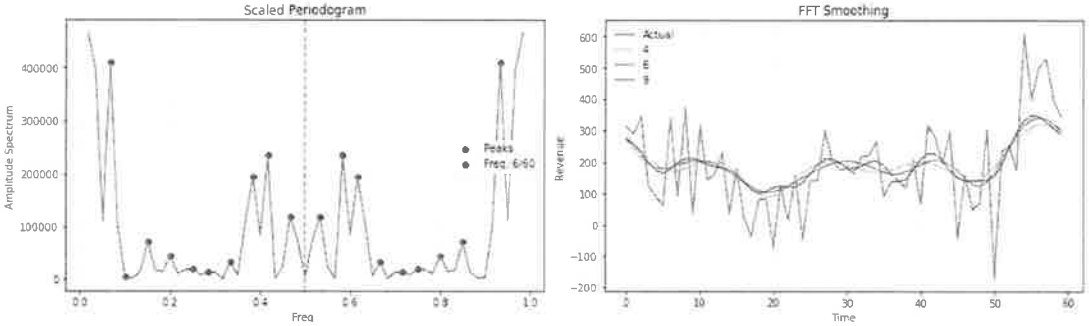
FX

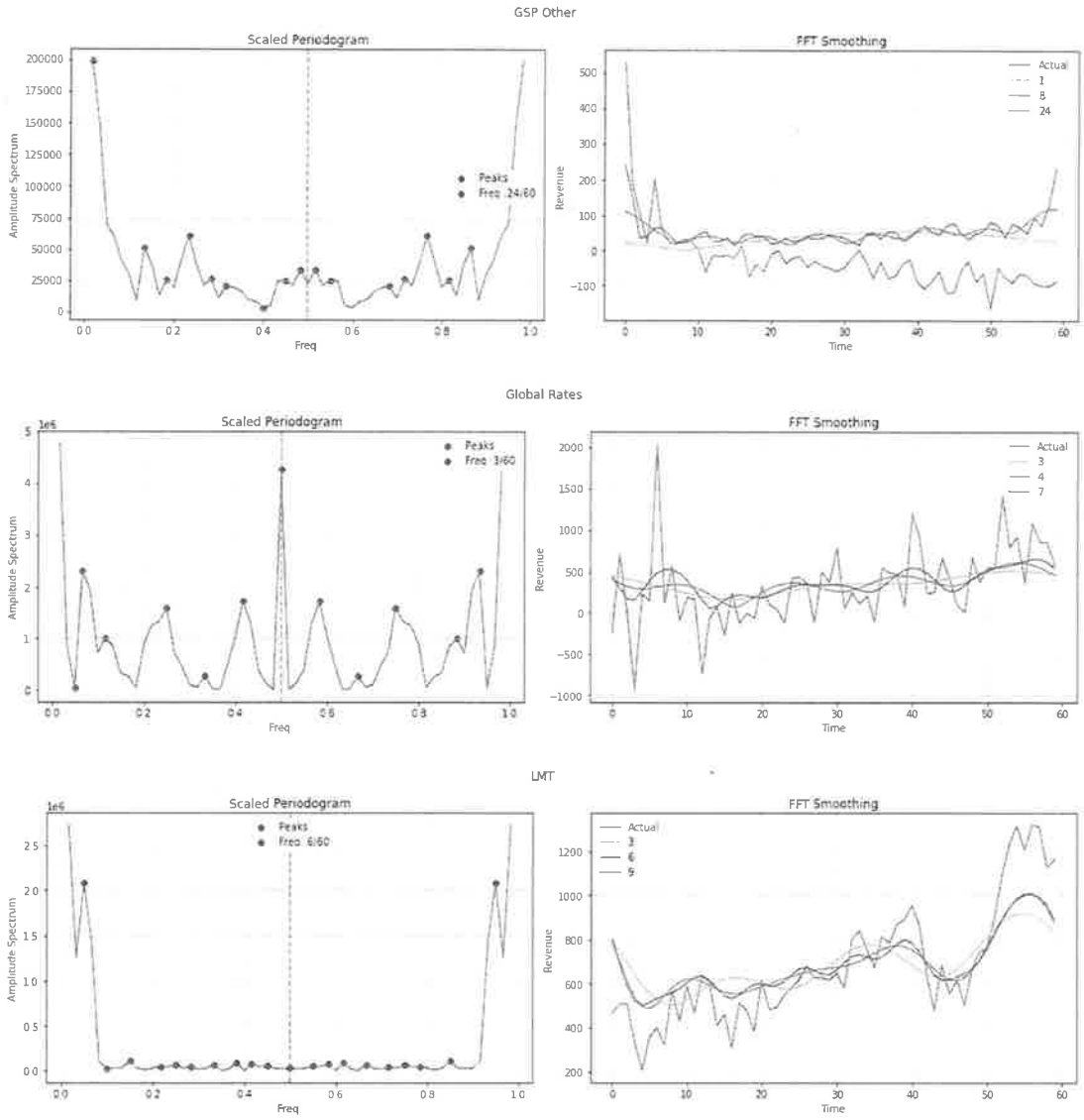


Finance Desk



G10 MT





See the body of the report for Prime Finance and Other periodograms.

Wavelet Intuition

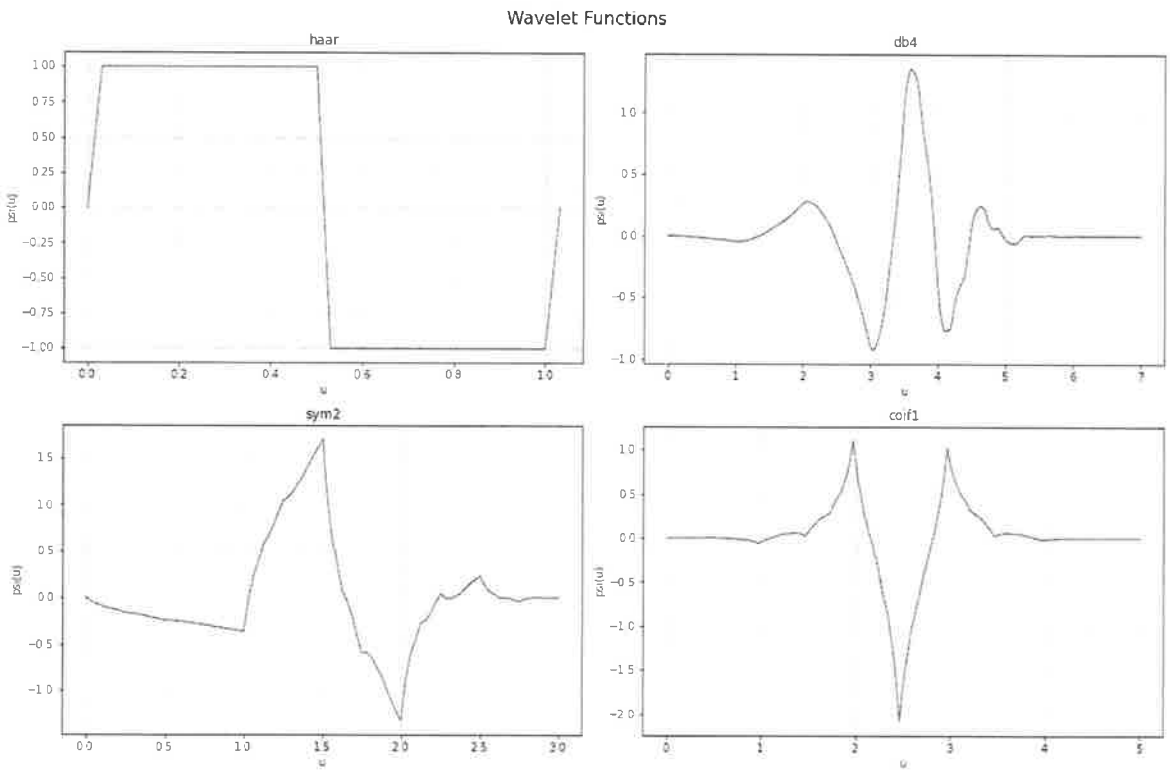
I outline some intuitions for the Wavelet transformation, beginning with the concept of scale. Let x(t) be a continuous time series, specifically a real valued function of t. Define the 'average' of x(t) over an interval [a,b] as:

$$\frac{1}{b-a}\int_a^b x(t)dt$$

Reparametrize the interval. Instead of set endpoints, used the width of the interval, λ, centered around t. Call λ the scale:

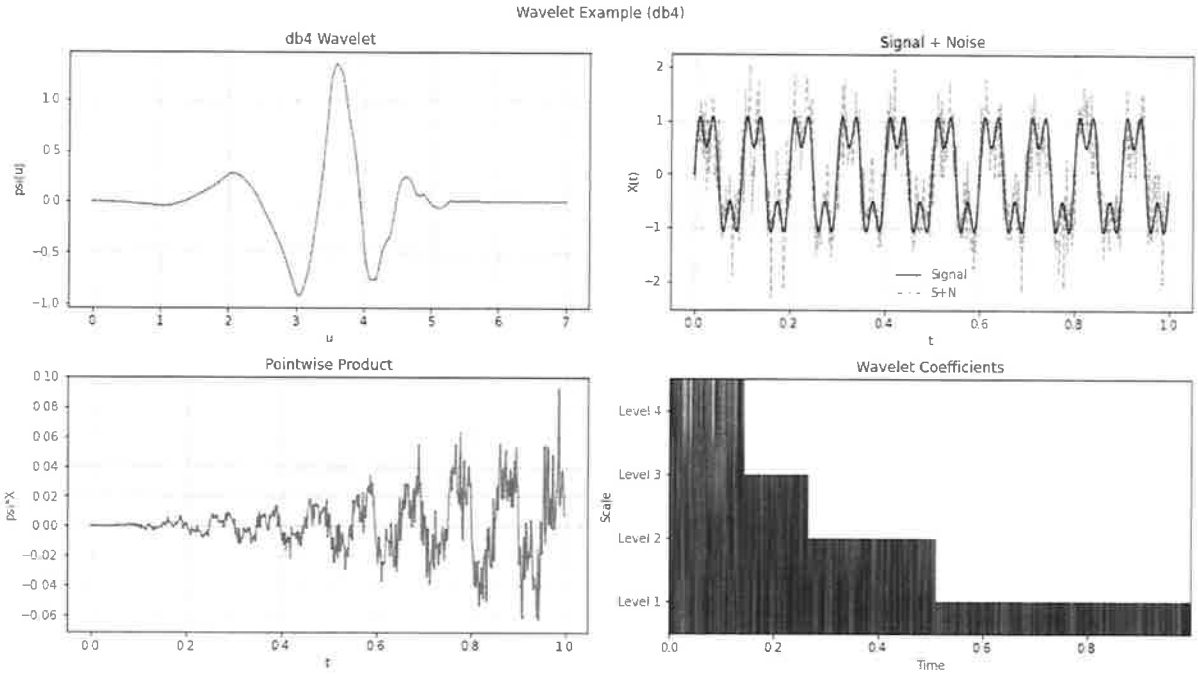
$$A(\lambda, t) \equiv \frac{1}{\lambda}\int_{t-0.5\lambda}^{t+0.5\lambda} x(u)du$$

And A(λ, t) the average value of x over scale λ centered at t. For example, A(λ,t) could be the average PnL of a trading desk over a one minute interval on a given day. Next, I provide examples of Haar, Daubechies, Symlet, and Coiflet wavelets:



I simulate a time series constructed from two sinusoids plus white noise and apply a db4 Wavelet transformation. Plot one (top left) shows the db4 wavelet. Plot two (top right) shows the series in the time domain. Plot 3 shows the pointwise product of the signal and wavelet function to capture the intuition of how the wavelet function interacts with the signal. I stretch the wavelet function to the same length as the noisy signal, hence around t = 0.0, the pointwise product shows small values, and where the signal is near zero, so is the product. The large spikes in the product are where the wavelet's main oscillations and the signal's oscillations coincide. Plot four (bottom right) shows the wavelet coefficients in scale-time space. Increasing decomposition levels corresponds to coarser scales (wider

snapshots of the signal in time). The color coding indicates magnitude of the Wavelet coefficient, with brighter colors indicating stronger correlation between wavelet and signal and darker, lower.



Trend Seasonality Regression Results

Trend-Seasonality Regression Results

Regression Output (3/3)	LMT	Other	Prime Finance
Q1	470.0726*** (68.2779)	51.1283 (130.2185)	128.2038*** (17.3907)
Q2	465.7705*** (60.1417)	-5.0454 (100.5832)	147.7535*** (19.1003)
Q3	442.1464*** (58.5259)	-57.0765 (97.5683)	137.6603*** (22.8647)
Q4	377.8644*** (54.6108)	-83.8690 (96.0165)	116.1488*** (19.4974)
T	-1.2845 (4.6149)	-0.9012 (6.9763)	7.7193*** (1.5506)
T^2	0.2258*** (0.0773)	0.0084 (0.1002)	-0.0869*** (0.0260)
R^2	0.7164	0.1474	0.5170
R^2 Adj.	0.6901	0.0684	0.4723

Regression output for market desks revenue trend-seasonality model: Rev(t) = B1*I(Q1) + ... + B4*I(Q4) + B5*t + B6*t^2 + e(t). Dependent variable is quarterly revenue in levels, non-seasonally adjusted, for Q1 2010 – Q4 2024, inclusive (T=60). Intercept omitted to avoid dummy trap. Newey-West standard errors in parenthesis, * p<.1, ** p<.05, ***p<.01, with a maximum of 4 lags. Lag order selected using AIC.

Lowess Cross-Validation Procedure

Notation:

- X: discrete time signal
- N: number of observations
- k: number of nearest observations
- f: fraction of observations, k/N
- i: index of observation x[i], 0....N-1

Algorithmically, the cross-validation procedure to select the optimal fraction of neighboring observations for Lowess smoothing is:

For each f in 0 ... 1.0:

    For i = 0 ... N-1:

        Let 'X modified' be X with x[i] removed

        Fit Lowess(f) to X modified

        Predict x[i] using interpolation between Lowess fitted values at x[i-1] and x[i+1]

        Compute the squared error between x[i] from X and predicted x[i]

    Average the squared errors over i

Select the f which minimizes the MSE

I use increments of 0.01 between the range 0.0 to 1.0, inclusive, as candidate fraction values. The CV procedure fits a regression for each observation and each possible fraction, meaning 101*N regressions.

Levels MEV Derivation

Estimating a first-differenced regression of the form:

$$\text{Rev}_t - \text{Rev}_{t-1} = \beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-1}) + e_t \ (1)$$

presumes that revenue in levels has a linear time trend:

$$\text{Rev}_t = \beta_0 t + \beta_1 \text{MEV}_t + u_t \ (2)$$

This is evident from assuming (2) and then first-differencing to obtain (1):

$$\begin{aligned}
\text{Rev}_t &= \beta_0 t + \beta_1 \text{MEV}_t + u_t \\
\text{Rev}_{t-1} &= \beta_0(t-1) + \beta_1 \text{MEV}_{t-1} + u_{t-1} \\
\text{Rev}_t - \text{Rev}_{t-1} &= \beta_0 t + \beta_1 \text{MEV}_t + u_t - \beta_0(t-1) - \beta_1 \text{MEV}_{t-1} - u_{t-1} \\
&= \beta_0 + \beta_0 t - \beta_0 t + \beta_1 \text{MEV}_t - \beta_1 \text{MEV}_{t-1} + u_t - u_{t-1} \\
&= \beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-1}) + e_t
\end{aligned}$$

where e(t) is defined as u(t) – u(t-1). (2) also follows from recursive substitution:

$$\begin{aligned}
\text{Rev}_t - \text{Rev}_{t-1} &= \beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-1}) + e_t \\
\text{Rev}_t &= \text{Rev}_{t-1} + \beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-1}) + e_t \\
&= \text{Rev}_{t-2} + \beta_0 + \beta_1(\text{MEV}_{t-1} - \text{MEV}_{t-2}) + e_{t-1} + \beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-1}) + e_t \\
&= \text{Rev}_{t-2} + 2\beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-2}) + e_{t-1} + e_t \\
&= \text{Rev}_{t-3} + \beta_0 + \beta_1(\text{MEV}_{t-2} - \text{MEV}_{t-3}) + e_{t-2} + 2\beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-2}) + e_{t-1} + e_t \\
&= \text{Rev}_{t-3} + 3\beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-3}) + e_{t-2} + e_{t-1} + e_t \\
&\vdots
\end{aligned}$$

Repeat the procedure t times to obtain:

$$\begin{aligned}
\text{Rev}_t &= \text{Rev}_0 + t\beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_{t-t}) + e_1 + \ldots + e_{t-2} + e_{t-1} + e_t \\
&= \text{Rev}_0 + t\beta_0 + \beta_1(\text{MEV}_t - \text{MEV}_0) + \sum_{j=1}^{t} e_j
\end{aligned}$$

Because a time-series is a time-indexed collection of random variables, substitution can continue recursively at t-1, t-2, ..., 0, -1, -2, to $-\infty$. However, modelers only observe a single realization of the process that begins at 0. To obtain a closed form solution, set:

$$\text{Rev}_0 \equiv \beta_1 \text{MEV}_0 + u_0$$

where u(t) is defined such that e(t) = u(t) – u(t-1):

$$
\begin{aligned}
\sum_{j=1}^{t} e_j &= \sum_{j=1}^{t} (u_j - u_{j-1}) \\
&= (u_1 - u_0) + (u_2 - u_1) + (u_3 - u_2) + \ldots + (u_{t-1} - u_{t-2}) + (u_t - u_{t-1}) \\
&= u_t - u_0 \\
u_0 &= u_t - \sum_{j=1}^{t} e_j
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Rev}_t &= \mathrm{Rev}_0 + t\beta_0 + \beta_1(\mathrm{MEV}_t - \mathrm{MEV}_0) + \sum_{j=1}^{t} e_j \\
&= \beta_1 \mathrm{MEV}_0 + u_t - \sum_{j=1}^{t} e_j + t\beta_0 + \beta_1(\mathrm{MEV}_t - \mathrm{MEV}_0) + \sum_{j=1}^{t} e_j \\
&= \beta_0 t + \beta_1 \mathrm{MEV}_t + u_t
\end{aligned}
$$

obviating Rev(0), or at any other value of t for that matter. Jump-off revenue is thus:

$$\widehat{\mathrm{Rev}}_{Q0} = \hat{\beta}_0 \cdot \mathrm{Q0} + \hat{\beta}_1 \mathrm{MEV}_{\mathrm{Q0}}$$

where the fitted coefficients come from OLS estimates of (1). Take the regression's fitted coefficients and multiply by the MEV actual, in levels, at Q_0.

HP Filter λ Estimation

Schlicht (2004) details a procedure to estimate the smoothing parameter λ in Hodrick-Prescott's filter. Hodrick-Prescott (1997) recommends λ values based upon the frequency of the time series to be smoothed, but many papers since present estimators for λ to derive its value empirically, from the underlying data. I implement Schlicht's maximum likelihood estimator, using the closed form expressions from the paper. I summarize the key results. For T observations of a series x, comprised of a trend component y and irregular component u, where v is the second differenced trend component:

$$x_{Tx1} = y_{Tx1} + u_{Tx1}$$
$$v = Py$$
$$P_{(T-2)xT} \equiv \begin{bmatrix} 1 & -2 & 1 & & & \dots & & 0 \\ 0 & 1 & -2 & 1 & & \dots & & 0 \\ & & & \dots & & & & \\ 0 & 0 & 0 & 0 & \dots & 1 & -2 & 1 \end{bmatrix}$$

Define α to be the smoothing parameter, and the estimates are:

$$\alpha = \frac{\sigma_u^2}{\sigma_v^2}$$
$$\hat{y} \equiv (I + \alpha P' P)x$$
$$\hat{u} \equiv x - \hat{y}$$
$$\hat{v} \equiv P\hat{y}$$

Let R(α) be a weighted average of the sum of squared estimation errors:

$$R(\alpha) = \hat{u}'\hat{u} + \alpha\hat{v}'\hat{v}$$

Despite pages of derivations[32], the likelihood function is straightforward to implement:

$$L(x : \alpha) = -\log\left(\det(I + \alpha P' P)\right) - T \cdot \log(R(\alpha)) + (T + 2) \cdot \log(\alpha)$$

I observe common issues with estimation. The optimal value is highly sensitive to the optimizer's starting point. I experimented with 1600 and the spline prescribed lambda and find the former traps the SLSQP algorithm in a local optimum. Alternative algorithms exhibit numerical stability issues and convergence failure. Most importantly, the reliability of the estimated smoothing parameter is contingent on a sample much larger than the T=60 available for revenue data.

[32] See Equation 85: https://www.econstor.eu/bitstream/10419/104132/1/lmu-mdp_2004-02.pdf

Baseline Smoothing Methods

Moving Average:

$$\widehat{\text{Rev}}_t = \frac{1}{k}\sum_{j=0}^{k-1} \text{Rev}_{t-j}$$

Smoothed revenue at period t is the average of revenue at t, t-1, ..., t-(k-1). The current jump off approach uses t=0 and k=4, thus averaging revenue at Q_0, Q_{-1}, Q_{-2}, Q_{-3}. Even if revenue is serially uncorrelated, the MA approach necessarily induces serial correlation in the residuals, where residuals equal actual minus smoothed. Assume that revenue is a white noise process:

$$\text{Rev}(t) \sim \text{WN}(0, \sigma_r^2)$$

The smoothed value of revenue at t is a function of actual revenue at t...t-(k-1), and smoothed revenue at t-i is a function of revenue at t-i, t-i-1,...,t-i-(k-1). There is no overlap for $i \geq k$, and the covariance is 0. For $i < k$, or equivalently $i = k - j$, $j \geq 1$, j terms overlap:

$$\text{Cov}(\widehat{\text{Rev}}_t, \widehat{\text{Rev}}_{t-i}) = \begin{cases} \frac{\sigma_r^2}{k} \text{ if } i = 0 \\ 0 \text{ if } i \geq k \\ \frac{j\sigma_r^2}{k^2} \text{ for } i = k - j,\ 1 \leq j \leq k \end{cases}$$

$$\text{Cov}(\widehat{\text{Rev}}_t, \text{Rev}_{t-i}) = \begin{cases} \frac{\sigma_r^2}{k} \text{ if } 0 \leq i \leq k-1 \\ 0 \text{ if } i \geq k \end{cases}$$

$$\text{Cov}(\text{Rev}_t, \text{Rev}_{t-i}) = \begin{cases} 0 \text{ if } i \neq 0 \\ \sigma_r^2 \text{ if } i = 0 \end{cases}$$

$$\text{Cov}(\text{Rev}_t, \widehat{\text{Rev}}_{t-i}) = \begin{cases} \frac{\sigma_r^2}{k} \text{ if } i = 0 \\ 0 \text{ if } i \geq 1 \end{cases}$$

Residual autocovariance is thus:

$$\begin{aligned} \text{Cov}(e_t, e_{t-i}) &= \text{Cov}(\text{Rev}_t - \widehat{\text{Rev}}_t, \text{Rev}_{t-i} - \widehat{\text{Rev}}_{t-i}) \\ &= \text{Cov}(\text{Rev}_t, \text{Rev}_{t-i}) - \text{Cov}(\text{Rev}_t, \widehat{\text{Rev}}_{t-i}) - \text{Cov}(\widehat{\text{Rev}}_t, \text{Rev}_{t-i}) + \text{Cov}(\widehat{\text{Rev}}_t, \widehat{\text{Rev}}_{t-i}) \\ &= \begin{cases} \sigma_r^2 - \frac{\sigma_r^2}{k} - \frac{\sigma_r^2}{k} + \frac{\sigma_r^2}{k} \text{ if } i = 0 \\ 0 - 0 - \frac{\sigma_r^2}{k} + \frac{(k-i)\sigma_r^2}{k^2} \text{ if } 1 \leq i \leq k-1 \\ 0 - 0 - 0 + 0 \text{ if } i \geq k \end{cases} \\ &= \begin{cases} \sigma_r^2 \left(\frac{k-1}{k}\right) \text{ if } i = 0 \\ \frac{-i\sigma_r^2}{k^2} \text{ if } 1 \leq i \leq k-1 \\ 0 \text{ if } i \geq k \end{cases} \end{aligned}$$

unlike an MA process in which autocovariance decays to 0, and is 0 after k lags, residual autocovariance is negative and *increasing* until k lags, after which it is 0. By construction, the MA smoothing procedure generates serially correlated residuals, contrary to the signal plus white noise smoothing framework. MA smoothed revenues uniformly exhibit greater serial correlation than actual revenues at lags 1-3. This is unsurprising given the overlap in underlying revenue actuals.

What is surprising is that the residuals from MA smoothing exhibit greater serial correlation than actuals for several desks, with the effect most pronounced at lag 3, the end of the MA window.

Difference in Serial Correlation, Actuals vs. MA Smoothed and Actuals vs. MA Smoothed Residuals

Desk	$\lvert\rho_{Actual}(h)\rvert - \lvert\rho_{MA}(h)\rvert$				$\lvert\rho_{Actual}(h)\rvert - \lvert\rho_{Residual}(h)\rvert$			
h	1	2	3	4	1	2	3	4
Commodities	-0.377	-0.395	-0.368	-0.248	0.508	0.283	0.075	0.444
DM - Credit Trading	-0.696	-0.682	-0.406	0.267	0.192	**-0.365**	**-0.177**	**-0.025**
DM - Structured Products Trading	-0.626	-0.693	-0.466	0.036	0.270	**-0.303**	**-0.128**	0.060
EMCT	-0.897	-0.519	-0.652	0.040	**-0.384**	0.233	-0.368	**-0.035**
Equity Admin	-0.577	-0.514	-0.509	-0.133	0.285	**-0.083**	**-0.109**	0.264
Equity Cash	-0.606	-0.586	-0.421	-0.111	0.209	0.101	0.198	**-0.041**
Equity Derivatives	-0.687	-0.552	-0.686	-0.074	0.032	0.221	**-0.284**	0.019
F&S	-0.240	-0.256	-0.253	-0.149	0.420	0.486	0.020	0.175
FDC - Clearing	-0.083	-0.118	-0.098	-0.059	0.750	0.520	0.600	0.697
FX	-0.551	-0.463	-0.367	-0.080	0.256	0.114	**-0.098**	0.067
Finance Desk	-0.237	-0.216	-0.184	-0.116	0.284	0.435	0.107	**-0.145**
G10 MT	-0.598	-0.248	-0.286	-0.262	0.093	0.266	0.191	**-0.186**
GSP Other	-0.246	-0.314	-0.353	-0.074	0.495	0.565	0.448	0.544
Global Rates	-0.685	-0.488	-0.500	-0.057	0.113	0.209	**-0.401**	**-0.018**
LMT	-0.085	-0.088	-0.086	-0.074	0.495	0.651	0.756	0.643
Other	-0.531	-0.718	-0.457	-0.009	0.283	**-0.373**	0.026	0.091
Prime Finance	-0.373	-0.396	-0.313	-0.220	0.582	0.250	0.319	0.412

Difference between desk revenue actuals absolute autocorrelation and MA smoothed revenue absolute autocorrelation (columns 2-5), and difference between desk revenue actuals absolute autocorrelation and MA smoothed revenue **residuals** absolute autocorrelation. MA smoothing performed as follows: Rev(t) = 0.25*(Rev(t) + Rev(t-1) + Rev(t-2) + Rev(t-3)). Autocorrelation computed at lags 1-4. Actuals span 2010 Q1 – 2024 Q4, inclusive (T=60). MA smoothed and residuals, where residuals = actual – smoothed, drop first 3 observations. A positive (negative) value indicates actuals show greater (less) absolute serial correlation than smoothed or residual.

MA smoothing for DMCT, DMSP, Equity Admin, Equity Derivatives, FX, and Global Rates induces a greater temporal dependence structure than exists in the actual revenue.

Cubic Splines

For observations $x_1 \ldots x_n$, a smoothing spline is an estimator that performs regularized regression over the basis of a natural spline. A spline is a piecewise continuous polynomial function, defined over intervals demarcated by knot points. A spline of degree k has a polynomial function of maximum degree k and continuous derivatives of order 1…k-1. The case of k=3 is a cubic spline, with continuous first and

second derivatives. A natural spline imposes the further restriction of the function to the left of the leftmost knot and to the right of the rightmost knot is of lower degree; for a cubic natural spline, said functions are linear. A smoothing spline uses all observations as knots, minimizing over all functions f:

$$\min_{f} \sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 dx$$

For λ a non-negative penalty parameter for smoothness. The summation term measures deviation from data. The second derivative of f(·) measures curvature. At λ=0, the optimal f(·) interpolates the data. At the limit for λ, the optimal f is linear as a linear function has second derivative 0. I use Scipy's implementation, which by default selects the smoothing parameter using generalized cross-validation. Smoothing splines are non-parametric approaches that produce visually appealing smoothed series on account of the continuity for the first and second derivatives. However, as is evident from the objective function, they do not incorporate any economic or time series structure. Splines do not account for autocorrelation, seasonality, or any other dependencies. For several desks, the spline's fitted value at Q_0 is quite close to actual revenue, so they provide little value for jump off analysis. Splines and other cross-sectional smoothing methods are better suited to a random walk signal than other time series methods.

Variable Score Weight Method Rankings

Adjusting weights for each component of the score function produces different rankings of methods. For revenue actuals y(t), denoised/smoothed g(t), and residuals e(t) = y(t) – g(t), weights correspond to:

Λ1: Serial Correlation of Residuals, measured by Ljung-Box Q statistic computed for e(t)

Λ2: Time-Varying Volatility of Residuals, measured by L2 norm of rolling volatility minus volatility

Λ3: CCF between y(t) and g(t)

Λ4: L1 norm of first-differenced y(t) minus first-differenced g(t)

Λ5: L2 norm of second differenced g(t)

I rank methods from 1st (best) to 10th (worst) per desk, and then aggregate ranks across desks. The HP Filter and FFT CV methods consistently perform well. Auto ARIMA and Trend/Seasonality regressions are often in the top 5, faring worse when the weights favor smoothness over residual properties.

Smoothing/Denoising Methods Rankings Per Weighting Scheme

Λ1	Λ2	Λ3	Λ4	Λ5	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
1.0	0.0	0.0	0.0	0.0	Auto Arima	HP Filter	Lowess	Reg.	FFTCV	Wavelet	FFT1	Hamilton	MA	Cubic Spline
0.0	1.0	0.0	0.0	0.0	Cubic Spline	MA	Lowess	Reg.	Wavelet	FFTCV	Hamilton	HP Filter	FFT1	Auto Arima
0.0	0.0	1.0	0.0	0.0	MA	Reg.	Lowess	Cubic Spline	Hamilton	FFTCV	FFT1	HP Filter	Wavelet	Auto Arima
0.0	0.0	0.0	1.0	0.0	HP Filter	FFTCV	FFT1	Lowess	Cubic Spline	Wavelet	MA	Hamilton	Auto Arima	Reg.
0.0	0.0	0.0	0.0	1.0	Hamilton	Auto Arima	Reg.	HP Filter	MA	Lowess	FFTCV	FFT1	Wavelet	Cubic Spline
0.2	0.2	0.20	0.20	0.20	HP Filter	FFTCV	Lowess	Wavelet	Cubic Spline	FFT1	MA	Hamilton	Reg.	Auto Arima
0.8	0.2	0.0	0.0	0.0	Auto Arima	HP Filter	Lowess	FFTCV	Reg.	Wavelet	FFT1	MA	Hamilton	Cubic Spline
0.7	0.3	0.0	0.0	0.0	Lowess	HP Filter	FFTCV	Auto Arima	Reg.	Wavelet	FFT1	MA	Hamilton	Cubic Spline
0.6	0.4	0.0	0.0	0.0	HP Filter	Lowess	FFTCV	Reg.	Auto Arima	Wavelet	MA	FFT1	Cubic Spline	Hamilton
0.5	0.5	0.0	0.0	0.0	HP Filter	Lowess	FFTCV	Reg.	Wavelet	Auto Arima	MA	Cubic Spline	Hamilton	FFT1
0.6	0.2	0.0	0.0	0.20	Auto Arima	Lowess	HP Filter	FFTCV	Reg.	Wavelet	Hamilton	MA	FFT1	Cubic Spline
0.0	0.0	0.40	0.40	0.20	HP Filter	FFTCV	FFT1	Lowess	Cubic Spline	Wavelet	Hamilton	MA	Reg.	Auto Arima
0.0	0.0	0.20	0.80	0.0	HP Filter	FFTCV	FFT1	Lowess	Cubic Spline	Wavelet	MA	Hamilton	Reg.	Auto Arima
0.0	0.0	0.50	0.50	0.0	HP Filter	FFTCV	FFT1	Lowess	Cubic Spline	Wavelet	MA	Hamilton	Reg.	Auto Arima
0.0	0.0	1/3	1/3	1/3	HP Filter	FFT1	FFTCV	Lowess	Wavelet	Cubic Spline	Hamilton	MA	Reg.	Auto Arima
0.4	0.3	0.10	0.10	0.10	FFTCV	Lowess	HP Filter	Wavelet	Reg.	Auto Arima	Cubic Spline	MA	FFT1	Hamilton

Lambda weights correspond to score components residual serial correlation, time-varying residual volatility, L1 norm of the difference in first-differenced series at inflection points, L2 norm of second differenced smoothed/denoised at local extrema, and CCF. Weights are non-negative and sum to 1. Lower scores are better. FFTCV denotes Fast Fourier Transform with cross-validation for cutoff frequency, FFT1 denotes 1st periodogram peak for cutoff frequency. MA uses 4Q boxcar kernel, HP Filter uses smoothing parameter of 1600. Reg. is a trend/seasonality regression of revenue on linear and quadratic time trends and quarterly indicators. Wavelet smoothing uses db4 and threshold 1.

Holdback Testing Results

The key premise of the jump-off framework is that some estimated Q_0 is a better representation of the business's revenues going forward than actual Q_0. Given a time series of actual revenue of each desk, I can test this premise and therefore compare the performance of smoothing methods explicitly by temporally splitting the actuals into a train and test set, fitting each method[33] to the train set, and using the last fitted observation in the training set as a static forecast for all periods in the test set

$$\text{Train: } \text{Rev}_{-(T-k-1)} \ldots \text{Rev}_0$$
$$\text{Test: } \text{Rev}_1 \ldots \text{Rev}_k$$
$$\overline{\text{Rev}}_{Q:1\ldots k} = \overline{\text{Rev}}_0$$

for T observations, and k equal to 4, 9, or 15, corresponding to standard FP&A forecasting horizons. I report smoothing method rankings across all desks using mean absolute percentage error[34] (lower is better) and R^2 (higher is better) for revenue in levels and first-differenced. Q_0 actual is the baseline.

Smoothing or denoising concerns drawing inferences about a true signal from data that was noisily measured in the past. The jump off framework *assumes* the true past signal is relevant for forecasts of the signal going forward. If the assumption holds, then the absolute predictive power of the true signal is not informative. I identify which denoising/smoothing method best reconstructs the true signal by extending the relevance assumption and comparing relative measures of predictive power.

Holdback Results, Levels Revenue

Ranking	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q: Train 2010 Q1 – 2023 Q4 (T=56), Test 2024 Q1 – 2024 Q4 (T=4)											
MAPE	MA	Lowess	Cubic Spline	Q_0 Actual	HP Filter	FFTCV	Auto Arima	Hamilton	Wavelet	Reg.	FFT1
R^2	MA	Cubic Spline	HP Filter	Lowess	FFTCV	Q_0 Actual	Wavelet	Hamilton	Auto Arima	Reg.	FFT1
9Q: Train 2010 Q1 – 2022 Q3 (T=51), Test 2022 Q4 – 2024 Q4 (T=9)											
MAPE	Reg.	MA	Lowess	HP Filter	FFTCV	Auto Arima	Q_0 Actual	Hamilton	Cubic Spline	FFT1	Wavelet
R^2	Reg.	HP Filter	FFTCV	Lowess	MA	Auto Arima	Q_0 Actual	Hamilton	Cubic Spline	FFT1	Wavelet
15Q: Train 2010 Q1 – 2021 Q1 (T=45), Test 2021 Q2 – 2024 Q4 (T=15)											
MAPE	FFTCV	Hamilton	Lowess	HP Filter	Cubic Spline	MA	Auto Arima	FFT1	Reg.	Q_0 Actual	Wavelet
R^2	FFTCV	HP Filter	Cubic Spline	Lowess	MA	Auto Arima	Hamilton	Q_0 Actual	Reg.	Wavelet	FFT1

Ranking of smoothing methods based on minimal mean absolute percentage error and maximal r-squared using Q0 from smoothing method fitted to revenue in levels, 2010 Q1 – 2023 Q4 (4Q), - 2022 Q3 (9Q), - 2021 Q1 (15Q) as predicted over subsequent 4, 9, and 15Q.

In levels, and across all desks, the best predictors on balance of the test 4Q, 9Q, and 15Q, are the 4Q moving average, trend-seasonality regression, and the FFT with cross-validation. The 4Q result is unsurprising because the average captures the local trend. Lowess is fundamentally a local regression

[33] Smoothing methods have no exposure to the test set, so I can safely include two-sided filters like the HP filter and signal processing methods.
[34] R^2 and MAPE produce different rankings because MAPE penalizes over forecasts more than under forecasts, whereas R^2 is a 1:1 function of MSE, which is symmetric. For example, forecasts of $50mm for actual of $25mm and $50mm for an actual of $75mm have the same squared error, but APE of the former is 100%, and the latter, 33%. Moreover, R^2 measures a reduction in variance relative to forecasting the mean.

and performs well for the same reason. At longer horizons, it is evident that FFT with CV's jump off better captures the go-forward state of the business. Testing first-differenced revenue confirms this.

Holdback Results, First-Differenced Revenue

Ranking	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q: Train 2010 Q2 – 2023 Q4 (T=55), Test 2024 Q1 – 2024 Q4 (T=4)											
MAPE	Lowess	FFTCV	HP Filter	MA	FFT1	Cubic Spline	Hamilton	Wavelet	Auto Arima	Reg.	Q_0 Actual
R^2	FFTCV	FFT1	MA	Lowess	HP Filter	Cubic Spline	Hamilton	Reg.	Auto Arima	Wavelet	Q_0 Actual
9Q: Train 2010 Q2 – 2022 Q3 (T=50), Test 2022 Q4 – 2024 Q4 (T=9)											
MAPE	FFTCV	Lowess	HP Filter	FFT1	Reg.	MA	Cubic Spline	Auto Arima	Hamilton	Wavelet	Q_0 Actual
R^2	FFTCV	Lowess	HP Filter	FFT1	Hamilton	Reg.	MA	Cubic Spline	Auto Arima	Wavelet	Q_0 Actual
15Q: Train 2010 Q2 – 2021 Q1 (T=44), Test 2021 Q2 – 2024 Q4 (T=15)											
MAPE	FFTCV	Lowess	HP Filter	FFT1	MA	Auto Arima	Hamilton	Cubic Spline	Wavelet	Reg.	Q_0 Actual
R^2	FFTCV	Lowess	HP Filter	FFT1	Auto Arima	Hamilton	MA	Wavelet	Cubic Spline	Reg.	Q_0 Actual

Ranking of smoothing methods based on minimal mean absolute percentage error and maximal r-squared using Q0 from smoothing method fitted to first-differenced revenue, 2010 Q2 – 2023 Q4 (4Q), - 2022 Q3 (9Q), - 2021 Q1 (15Q) as predicted over subsequent 4, 9, and 15Q.

Differencing removes any linear time trend. Results show the relative predictive power of each method's estimate of Q_0 over different test horizons, as a proxy for the ability to remove noise from actual Q_0 and produce a jump-off better representative of the desk's go forward state. Absolute power is weak.

Rolling Testing Results

Holdback results using a single test set. Results are therefore sensitive to idiosyncrasies of the last 4, 9 and 15Q, spanning 2024 Q1, 2022 Q4, and 2021 Q2 to 2024 Q4, respectively. I use a rolling window back testing framework to get distributions of forecast errors per desk, smoothing/denoising method, and forecast horizon. Let h = 4, 9, 15 be the forecast horizon, d be the desk (e.g. Commodities, FX, Global Rates), m be the smoothing method (e.g. Lowess, Auto SARIMA, Hamilton Filter), T be the total sample size (60 quarters), W the rolling window length, and n a subsample of the data:

- h: forecast horizon, 4, 9, or 15Q ahead[35]
- d: desk
- m: smoothing/denoising method
- T: 60, the total set of quarterly revenues spanning 2010 Q1 – 2024 Q4, inclusive
- W: 32, the rolling window length[36]
- n: subsample of the data of length W. For 1Q window increments, there are n=1...T – W + 1 subsamples, e.g. 1...W, 2...W+1, 3...W+2, ... , T-W...T-1, T-W+1...T

Algorithmically, the procedure is as follows:

For each subsample length W=32, indexed 1...T-W-1

For each desk's revenue actuals

Select CV parameters from whole sample fit[37]

Apply smoothing/denoising methods to actuals in window

Retain fitted Q0 as a static forecast

For forecast period h=4, 9, 15

Compute forecast R2 and MAPE using fitted Q0 as predicted and actuals

Define error measures as follows, where index j = 1....h:

$$e_{d,n,m,j} \equiv \mathrm{Rev}_{d,n,j} - \widehat{\mathrm{Rev}}_{d,n,m,j},\ j = 1...h$$

$$R^2_{d,n,m,h} = 1 - \frac{\sum_{j=1}^{h} e^2_{d,n,m,j}}{\sum_{j=1}^{h} (\mathrm{Rev}_{d,n,j} - \overline{\mathrm{Rev}}_{d,n,h})^2}$$

$$\mathrm{MAPE}_{d,n,m,h} = \frac{1}{h} \sum_{j=1}^{h} \left| \frac{e_{d,n,m,j}}{\mathrm{Rev}_{d,n,j}} \right|$$

[35] Forecast horizons follow convention for PPNR markets models. The longer horizons reduce the number of subsamples. At h=4, the last rolling window ends at T=56, for h=9, at T=51, and for h=15, at T=45.

[36] The window size should be large enough to provide sufficient statistical power for smoothing methods, but small enough to produce many subsamples with variation in subsample forecasting error. I use W=32 accordingly. SARIMA models with high lag orders already trim observations when fit using MLE, so 8 years of quarterly data should suffice. Signal decomposition methods also need large samples to parse high frequency/low scale components. 32 is also a multiple of the 4-quarter period, thus accounting for intra-sample seasonality.

[37] Lowess and FFT require hyperparameters set using cross-validation. It is time-prohibitive to use cross-validation for each pass, hence I use the optimal parameters computed per desk over the complete T=-60 dataset. Lowess already requires 101*T regression fits for data of length T. At T=32 (window size), and maximum of 56-32+1=25 fits, this is 101*32*25 = 80,800 regressions.

Per forecast horizon, desk, and smoothing method, I take the median of error measures computed over subsamples. Preliminary analysis shows large outlier errors, hence the median instead of an average. I rank order the methods per horizon, aggregating over desks, favoring lower MAPE and higher R^2, and then rank again, aggregating over horizons. I repeat the exercise on first-differenced revenue, and report rankings for MAPE and R^2 separately. The latter shows reduction in forecast variance relative to the using the forecast period mean. Q_0 Actual denotes actual revenue at Q_0 as a baseline predictor:

Smoothing Method Rankings by Rolling MAPE (Revenue in Levels)

h	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q	**MA**	**HP Filter**	**Lowess**	**Cubic Spline**	**Hamilton**	**Reg.**	**Auto Arima**	**Wavelet**	**Q_0 Actual**	**FFT1**	**FFTCV**
9Q	**HP Filter**	**Hamilton**	**Auto Arima**	**MA**	**Lowess**	**Cubic Spline**	**Q_0 Actual**	**FFT1**	**Wavelet**	**FFTCV**	**Reg.**
15Q	**MA**	**Auto Arima**	**HP Filter**	**Cubic Spline**	**Lowess**	**Wavelet**	**Hamilton**	**FFTCV**	**Q_0 Actual**	**FFT1**	**Reg.**
Agg.	**MA**	**HP Filter**	**Lowess**	**Hamilton**	**Cubic Spline**	**Auto Arima**	**Wavelet**	**Q_0 Actual**	**FFT1**	**FFTCV**	**Reg.**

Ranking of methods aggregated over desks and horizon, using median MAPE computed over 25, 20, and 14 rolling subsamples for 4, 9, and 15Q horizons, respectively. W=32, T=60, Q1 2010 – Q4 2024 revenue actuals in levels, inclusive. Lower MAPE is better.

Smoothing Method Rankings by Rolling R^2 (Revenue in Levels)

h	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q	**HP Filter**	**MA**	**Lowess**	**FFT1**	**Cubic Spline**	**Wavelet**	**Hamilton**	**Q_0 Actual**	**FFTCV**	**Reg.**	**Auto Arima**
9Q	**HP Filter**	**MA**	**Lowess**	**Cubic Spline**	**Hamilton**	**FFT1**	**Reg.**	**Q_0 Actual**	**FFTCV**	**Wavelet**	**Auto Arima**
15Q	**HP Filter**	**MA**	**Cubic Spline**	**Lowess**	**Auto Arima**	**Hamilton**	**Reg.**	**Q_0 Actual**	**Wavelet**	**FFTCV**	**FFT1**
Agg.	**HP Filter**	**MA**	**Cubic Spline**	**Lowess**	**Hamilton**	**FFT1**	**Reg.**	**Q_0 Actual**	**Wavelet**	**FFTCV**	**Auto Arima**

Ranking of methods aggregated over desks and horizon, using median R2 computed over 25, 20, and 14 rolling subsamples for 4, 9, and 15Q horizons, respectively. W=32, T=60, Q1 2010 – Q4 2024 revenue actuals in levels, inclusive. Higher R2 is better.

MAPE rankings favor the moving average approach because it captures the local trend in revenues. Lowess and Cubic Spline perform well for the same reason, they draw on local features of the time series. Denoising methods that perform well in the complete sample, such as Wavelet, FFT with cross-validation, and Auto SARIMA, produce forecasts with high variance owing to the small sample of the W=32 quarter window. By R^2, they perform worse than predicting ahead using the last actual from the training window. At 15Q. Auto ARIMA does well in levels, measured by MAPE, by producing linear forecasts. The order of rankings differs using R^2 versus MAPE. FFT1 has high MAPE at 4Q, for example, because although it's short-term forecasts co-move with actuals, and thus it has relatively strong R^2 performance, the forecasts are at the wrong level. The HP filter is ranked first using R^2.

First-differenced results favor the FFT, HP Filter, FFT with cross-validation, and MA methods. First-differencing produces better measures of the predictive power of each smoothing/denoising method's fitted Q0 over the forecast horizon, as it eliminates linear time trends in the revenue actuals. Despite the small samples, the FFT can still isolate relevant signals. FFT1 outperforms FFTCV, as measured by MAPE, because it uses periodogram peaks which are much more informative in compressed samples. The regression model performs the worst because it is specified for data in levels. MA, Wavelet, and HP Filter produce flat forecasts, hence the low MAPE. Rolling results are consistent with holdback results.

Smoothing Method Rankings by Rolling MAPE (Revenue First-Differenced)

h	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q	**FFT1**	**FFTCV**	**MA**	**Lowess**	**HP Filter**	**Wavelet**	**Q0 Actual**	**Cubic Spline**	**Hamilton**	**Reg.**	**Auto Arima**
9Q	**FFT1**	**FFTCV**	**HP Filter**	**MA**	**Wavelet**	**Cubic Spline**	**Lowess**	**Hamilton**	**Q0 Actual**	**Auto Arima**	**Reg.**
15Q	**FFT1**	**FFTCV**	**HP Filter**	**MA**	**Wavelet**	**Cubic Spline**	**Lowess**	**Hamilton**	**Q0 Actual**	**Auto Arima**	**Reg.**
Agg.	**FFT1**	**FFTCV**	**HP Filter**	**MA**	**Wavelet**	**Lowess**	**Cubic Spline**	**Q0 Actual**	**Hamilton**	**Auto Arima**	**Reg.**

Ranking of smoothing/denoising methods aggregated across desks, per forecasting horizon and aggregated over desks and horizon, using median MAPE computed over 24, 19, and 13 rolling subsamples for 4, 9, and 15Q horizons, respectively. W=32, T=59, Q2 2010 – Q4 2024 revenue actuals in levels, inclusive. Lower MAPE is better.

Smoothing Method Rankings by Rolling R^2 (Revenue First-Differenced)

h	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
4Q	**FFT1**	**Q0 Actual**	**MA**	**HP Filter**	**Cubic Spline**	**Reg.**	**Lowess**	**Auto Arima**	**Hamilton**	**Wavelet**	**FFTCV**
9Q	**FFT1**	**HP Filter**	**MA**	**Hamilton**	**Lowess**	**Cubic Spline**	**Auto Arima**	**FFTCV**	**Q0 Actual**	**Wavelet**	**Reg.**
15Q	**FFT1**	**MA**	**FFTCV**	**Hamilton**	**HP Filter**	**Wavelet**	**Cubic Spline**	**Auto Arima**	**Lowess**	**Q0 Actual**	**Reg.**
Agg.	**FFT1**	**MA**	**HP Filter**	**Cubic Spline**	**Hamilton**	**Q0 Actual**	**Auto Arima**	**Lowess**	**FFTCV**	**Wavelet**	**Reg.**

Ranking of smoothing/denoising methods aggregated across desks, per forecasting horizon and aggregated over desks and horizon, using median R2 computed over 24, 19, and 13 rolling subsamples for 4, 9, and 15Q horizons, respectively. W=32, T=60, Q2 2010 – Q4 2024 revenue actuals in levels, inclusive. Higher R2 is better.

The small training sample size for each window and overlap in test sets preclude consideration of the rolling test results as determinative. Rolling tests are better suited to monthly data, especially with non-overlapping windows. 15 years of monthly data provides 180 observations, enough to segment test sets.

Extension: Smoothed Actuals for Regression

Separating signal from noise in historical revenue actuals is useful beyond choosing an appropriate jump-off value. PPNR markets models use transformed revenue (differenced, log differenced, net/gross percentage) as the dependent variable in regression analysis. Covariates are transformed macroeconomic variables. The regressions establish a contemporaneous relationship between the change in revenue and the change in macroeconomic variables, which enables scenario-based forecasting by setting the covariates equal to their scenario specific values. I conduct a simulation study to examine the effect of revenue smoothing on statistical significance and goodness of fit in regression analysis and complement the study with empirical examples from G10 MT and LMT.

The simulation study proceeds as follows. I simulate data by drawing from a completely specified process, then add noise and estimate the parameters of the noisy and noiseless process. I use many draws to compute empirical sample moments and compare the sampling distribution to theorized moments of the noiseless and noisy estimator. The noiseless data is the case of perfect smoothing because the true data and data-generating process is known. I derive complete results for the moments of the estimators. Let (1) be the true relationship and (2) an approximation thereof due to noise:

$$Y_{n \times 1} = X_{n \times k}\beta_{k \times 1} + \epsilon_{n \times 1} \quad (1)$$
$$Y_{n \times 1} + \delta_{n \times 1} \approx X_{n \times k}\beta_{k \times 1} + \epsilon_{n \times 1} \quad (2)$$

Assume noise delta is homoscedastic, and correlated with regression error:

$$\delta \sim (\vec{0}, \sigma_\delta^2 I)$$
$$\Sigma_{\epsilon\delta} = \Sigma_{\delta\epsilon} = \rho I_{n \times n}$$

Minimizing (2)'s sum of squared residuals produces a familiar form:

$$\begin{aligned}\hat{\beta}_{\text{OLS}}^{\text{noise}} &= (X'X)^{-1}(X'Y + X'\delta) \\ &= (X'X)^{-1}(X'Y) + (X'X)^{-1}(X'\delta) \\ &= \hat{\beta}_{\text{OLS}} + (X'X)^{-1}(X'\delta)\end{aligned}$$

This estimator is unbiased, with variance:

$$\begin{aligned}\text{Var}(\hat{\beta}_{\text{OLS}}^{\text{noise}} | X) &= (\sigma_\epsilon^2 + \sigma_\delta^2)(X'X)^{-1} + (X'X)^{-1}X'\rho IX(X'X)^{-1} + (X'X)^{-1}X'\rho IX(X'X)^{-1} \\ &= (\sigma_\epsilon^2 + \sigma_\delta^2 + 2\rho)(X'X)^{-1}\end{aligned}$$

If noisy actuals for the dependent variable manifest as higher variance of the error term, σ_e^2 but the error remains mean zero, the estimator is unbiased but with higher variance. Standard errors increase, T-statistics are lower, and p-values are higher. See below for simulation results that compare theoretical versus sampling distribution moments for the estimators of noiseless and noisy data.

Simulation Results, Noiseless Y(t)

Population		$E(\hat{\beta})$	$Cov(\hat{\beta})$		% of Estimates Significant at 5%
β_1	0.2	0.02003	0.0004423	0.001148	-
β_2	0.15	0.0004423	0.05578	-0.001929	
β_3	-0.9	0.001148	0.001929	0.06004	
Sample		$\bar{\hat{\beta}}$	$\widehat{Cov}(\hat{\beta})$		% of Estimates Significant at 5%

β_1	0.200191	0.02042	0.000141	0.001031	28.72%
β_2	0.155756	0.000141	0.05570	-0.002287	9.93%
β_3	-0.900625	0.001031	-0.002287	0.06020	95.03%

Population moments of estimator for noiseless Y(t) compared to sample mean and covariance of estimated parameters computed over repeated draws using N = 10,000 simulations of length T=50 from the process Y(t) = 0.2+0.15*X(1)-0.9*X(2) + e(t), e(t) ~ WN(0,1). Noise d(t) ~ WN(0,1) with correlation 0.6 with e(t). Noiseless data is Y(t), noise data is Y(t) + d(t). Percent of simulations for which regression coefficient estimates are significant at a 5% level are in the last column.

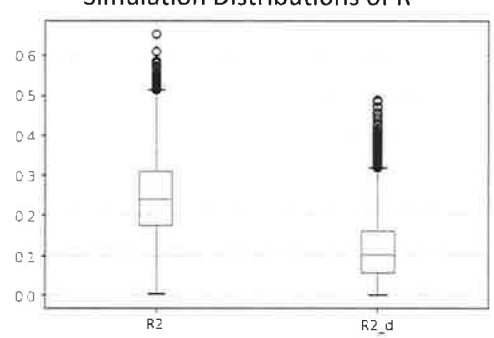
Simulation Results, Noisy Y(t) + d(t)

Population	$E(\hat{\beta})$	$Cov(\hat{\beta})$			% of Estimates Significant at 5%
β_1^d	0.2	0.06408346	0.00141543	0.003674	-
β_2^d	0.15	0.00141543	0.17850902	-0.006173	
β_3^d	-0.9	0.00367438	-0.0061728	0.1921	
Sample	$\bar{\hat{\beta}}$	$\widehat{Cov}(\hat{\beta})$			% of Estimates Significant at 5%
β_1^d	0.200692	0.06478	-0.000841	0.003558	12.43%
β_2^d	0.161248	-0.000841	0.17684	-0.008387	6.63%
β_3^d	-0.903522	0.003558	-0.008387	0.1933	52.07%

Population moments of estimator for noisy Y(t)+d(t) compared to sample mean and covariance of estimated parameters computed over repeated draws using N = 10,000 simulations of length T=50 from the process Y(t) = 0.2+0.15*X(1)-0.9*X(2) + e(t), e(t) ~ N(0,1). Noise d(t) ~ N(0,1) with correlation 0.6 with e(t). Noiseless data is Y(t), noise data is Y(t) + d(t). Percent of simulations for which regression coefficient estimates are significant at a 5% level are in the last column.

Sample averages and covariances largely match population moments. For each coefficient, a much lower percentage of simulations with noisy data produce statistically significant coefficient estimates at a 5% level. Noise thus obscures the true relationship. R^2 values are lower with noisy data:

Simulation Distributions of R^2



Box and whisker plots for R^2 values computed for each simulated noiseless Y(t) and noisy Y(t)+d(t). N=10,000 simulations of process Y(t) = 0.2+0.15*X(1)-0.9*X(2) + e(t), e(t) ~ N(0,1). Noise d(t) ~ N(0,1) with correlation 0.6 with e(t). Minimum values are 0.0, maximums are 0.653 and 0.487, respectively, with outliers marked using P75 + 1.5*IQR.

The simulation results show the following. First, even with noise added to the dependent variable, regression parameter estimates are unbiased if noise averages to 0. For pure measurement error, this is plausible, but noise can be systematically different from 0. Second, noise does increase the variance of parameter estimates. A lower proportion of simulations have statistically significant estimates at 5%, for each parameter, indicating statistically significant relationships can be obscured by noise. Third, regressions with noise tend to exhibit lower R^2. The intuition for this empirical result is that the noisy series has higher variance and therefore all else equal, the same model will explain less of the variation in the dependent variable. These results are unsurprising but require simulation to show conclusively.

Finally, I compare the fitted markets, rates, and NIR benchmark model and the Local Markets Treasury model to the same models fitted with smoothed revenue in levels, using 2, 4, 6, and 8Q moving averages and Lowess smoothing. The benchmark model uses a gross percentage change of revenue and the LMT model is in levels. LMT uses the OIS Annual vs. SOFR 2Y swap rate and the Citi World Government Bond Index as covariates.

Benchmark Fitted Model

$$\frac{\text{IndustryMacroFeeSA}_t}{\text{IndustryMacroFeeSA}_{t-1}} = 1.005319787 + 0.004021866\,(\text{Log3MSwaptionVol30YExpiry}_{t-1} - \text{Log3MSwaptionVol30YExpiry}_{t-2}) + 0.326949492\left(\frac{\text{JPMCurrencyVolIndexG7})_t}{\text{JPMCurrencyVolIndexG7})_{t-1}} - 1\right) + \epsilon_t$$

$$\hat{y}_t = 1.005319787 + 0.004021866\,(\text{Log3MSwaptionVol30YExpiry}_{t-1} - \text{Log3MSwaptionVol30YExpiry}_{t-2}) + 0.326949492\left(\frac{\text{JPMCurrencyVolIndexG7})_t}{\text{JPMCurrencyVolIndexG7})_{t-1}} - 1\right)$$

Unlike the simulation study, the true data generating process for benchmark fees is unknown. For comparison, I treat the model fitted to unsmoothed revenues as correct. I report regression estimates for 4Q, 8Q, and Lowess smoothed industry fees, using the same semiparametric form, below:

Benchmark Model Smoothed Revenue Regression Estimates

IndustryMacroFeeSA(t) Smoothing	None	4QMA	8QMA	Lowess(0.15)	Lowess(0.20)
Intercept	1.005*** (0.010)	1.001*** (0.010)	1.001*** (0.009)	1.001*** (0.008)	1.003*** (0.007)
G7_Currency_Vol(t)/G7_Currency_Vol(t-1) - 1	0.327*** (0.067)	0.136*** (0.038)	0.037 (0.027)	0.067*** (0.017)	0.019 0.019
3M30Y_Swap_Log_IV(t-1)- 3M30Y_Swap_Log_IV(t-2)	0.004*** (0.001)	0.002*** (0.000)	0.001** (0.000)	0.001*** (0.000)	0.000* (0.000)
R^2	0.259	0.277	0.090	0.149	0.028
R^2 Adj.	0.237	0.254	0.060	0.124	-0.001
T	70	67	63	70	70

Coefficient estimates, statistical significance, and goodness of fit for regressions of unsmoothed (None) and smoothed industry fees (4QMA, 8QMA, Lowess), transformed using a gross percentage, on the percentage change in currency volatility and first-differenced and lagged swaption implied volatility. 2006 Q2 – 2023 Q4 quarterly data, inclusive (T=70). Newey-West standard errors with 4 lags in parenthesis, ***:p<0.01, **:p<0.05, *:p<0.10. 4QMA smoothing omits first 3 observations, 8QMA omits first 7 observations.

More quarters used in the MA and higher Lowess fractions result in greater smoothing. Increased smoothing can reduce goodness of fit[38], alters parameter estimates, and degrades statistical significance. The distortionary effect of smoothing is attributable to the large spikes in benchmark fees around Covid and the GFC. Smoothing pushes those spikes temporally farther past the dates at which fees and volatility, forward and implied, co-move, thus destroying the contemporaneous relationship.

[38] R^2 increases marginally with 4QMA smoothing, although the effect is ambiguous owing to reduced sample size.

LMT Fitted Model

$$\text{Revenue}_t^{\text{LMT}} = 318,284,335.545 - 52,385,168.806 \cdot \ln(2\text{yRFRSwp}_t/2\text{yRFRSwp}_{t-1}) - 823,215,388.269 \cdot \ln(\text{SBWGU}_t/\text{SBWGU}_{t-1}) + u_t$$

$$\widehat{\text{Revenue}}_t^{\text{LMT}} = 318,284,335.545 - 52,385,168.806 \cdot \ln(2\text{yRFRSwp}_t/2\text{yRFRSwp}_{t-1}) - 823,215,388.269 \cdot \ln(\text{SBWGU}_t/\text{SBWGU}_{t-1})$$

More aggressive smoothing lowers R^2 for the LMT model. This is a counter-intuitive result, but summary statistics for the residuals, defined as actual minus smoothed, show bias increasing with the amount of smoothing. Smoothing in levels induces greater noise. MA smoothing is especially problematic in the small sample case and should be avoided for regression analysis. See below for estimates.

Local Markets Treasury Smoothed Revenue Regression Estimates

Smoothing	None	4QMA	8QMA	Lowess(0.15)	Lowess(0.2)
Intercept	318,284,335.545*** (12,405,535.943)	316,718,580.714*** (11,183,065.575)	314,951,906.004*** (7,376,205.360)	317,912,119.061*** (12,042,933.763)	316,067,761.929*** (11,771,405.253)
US_IR_2Y_RFRSWP_LnD1	-52,385,168.806*** (5,241,276.115)	-21,875,155.050*** (6,319,575.471)	-8,014,570.145*** (2,361,542.136)	-29,016,968.705*** (3,359,686.159)	-20,360,733.436*** (2,820,598.235)
US_EQ_SBWGU_LnD1	-823,215,388.269*** (189,983,058.016)	55,114,005.262 (160,115,076.745)	113,116,065.311 (82,546,702.052)	-430,137,888.049*** (158,314,502.683)	-234,543,796.293 (147,907,089.673)
R-squared	0.397	0.166	0.134	0.223	0.131
R-squared Adj.	0.355	0.102	0.055	0.169	0.071
T	32	29	25	32	32

Coefficient estimates, statistical significance, and goodness of fit for regressions of unsmoothed (None) and smoothed LMT revenue (4QMA, 8QMA, Lowess), on log differenced OIS vs. SOFR 2-year swap rate and first-differenced and lagged swaption implied volatility. 2016 Q1 – 2023 Q4 quarterly data, inclusive (T=32). Newey-West standard errors with 4 lags in parenthesis, ***:p<0.01, **:p<0.05, *:p<0.10. 4QMA smoothing omits first 3 observations, 8QMA omits first 7 observations.

The attached file contains complete derivations and results.

PPNR_Actuals_Smoothing.html

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